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Parametric and modal work-holding apparatus

Abstract

A system for inspecting each workpiece of a plurality of non-identical workpieces, each workpiece having a distinct workholding specification. The system includes a workholder configured to autonomously execute a plurality of workholding operations pursuant to parameters of a specified pre-defined workholding mode specified from a plurality of distinct, pre-defined workholding modes. Each pre-defined workholding mode specifies a plurality of holding parameters corresponding to the holding specification of a corresponding workpiece.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. non-provisional application Ser. No. 17/535,104, filed Nov. 24, 2021 and titled “Parametric and Modal Work-holding Method for Automated Inspection” and naming Jonathan J. O'Hare and Jonathan Dove as inventors; and is a continuation-in-part of U.S. non-provisional application Ser. No. 17/949,940, filed Sep. 21, 2022 and titled “Automated Work-holding for Precise Fastening of Light Parts during Automated Transfer” and naming Jonathan J. O'Hare and Jonathan Dove as inventors. (2) The disclosures of all of the foregoing are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

(1) Embodiments generally relate to production processes, more particularly, embodiments relate to inspection operations.

BACKGROUND ART

(2) One of the most rapidly growing areas in manufacturing is automation. Companies today need to be globally competitive and thus must be able to justify highly skilled labor through the efficiency of their operation. To this end, collaborative robots (COBOTS) as well as other automated machinery, must be effectively integrated into each production process and work as independently of human intervention as possible.

(3) One such production process in many manufacturing operations is the inspection or measurement process. Coordinate measuring machines (CMMs) have long been used to assist in providing critical measurement data to provide the necessary feedback to control all of the other processes responsible for producing the product. Conventional CMMs do not collaborate with other equipment or share the information they acquire to enable process level decisions to be made on their own. CMMs still often rely on human operators to make decisions to prepare parts for inspection as well as analyze the results for corrective action.

(4) Work-holding systems and methods for the inspection process often include hardware which is referred to as fixtures for specific types of manufactured parts. Any manufactured part that has work performed on it as part of a process may also more generally be referred to as a workpiece. It is important however to make the distinction between a work-holding method and a fixture, as the means by which to hold a workpiece for some operation to be performed on it may not include a fixture at all. In many cases, work-holding is accomplished through the use of a variety of common hardware components, such as screw clamps, vises, spring levers, mounting plates, riser blocks,

clay, glue, etc. The reason for this is that many inspection processes implement automated inspection systems which are intended to inspect a variety of workpieces, whereas in the case of other manufacturing operations, they are typically setup for a dedicated operation on a specific workpiece. It is therefore a common practice in most inspection processes to have a flexible means by which to hold different workpieces. One solution for this is to have dedicated fixtures designed for each workpiece, however this can be also extremely expensive since they must also be designed specifically for measurement sensor accessibility. Another common solution in industry is to have assortment of work-holding hardware in the form of modular fixture kits so that custom fixtures can be built on an as-need basis. The problem with using modular fixture kits as a work-holding method is that it may be inconsistently rebuilt over time causing reliability and measurement reproducibility problems. This is due to the fact that high accuracy measurements are often subject to deflection in the work-piece due to applied forces during work-holding. On the other extreme, a workpiece may not be held tightly enough so that there is unwanted or 'lost' motion during motion thereby causing false measurements. Lastly, the problem with both of aforementioned industry work-holding solutions is that they must be frequently switched out on the same inspection system that is being used for a variety of workpieces. The frequent changing of fixtures or work-holding methods between jobs on the same machine becomes an operational efficiency problem.

SUMMARY OF VARIOUS EMBODIMENTS

(5) In accordance with one embodiment, a method of operating a workholder during sequential inspection of a plurality of workpieces, each workpiece having a holding specification distinct from the respective holding specifications of other workpieces of the plurality of workpieces, includes providing a plurality of distinct, selectable pre-defined modes, each such mode specifying a plurality of holding parameters corresponding to the holding specification of a corresponding workpiece of the plurality of workpieces, each holding parameter of the plurality of holding parameters corresponding to a corresponding workholding operation of a plurality of workholding operations. The method also includes providing the workholder to hold each workpiece during inspection by an inspection instrument of an inspection system, the workholder configured to autonomously execute workholding operations pursuant as specified by the holding parameters of each of the pre-defined modes; and receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes, said pre-defined mode being a specified pre-defined mode. The method also includes causing the workholder to autonomously execute the plurality of workholding operations pursuant to the parameters of the specified pre-defined mode.

(6) In some embodiments, the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes includes receiving a set of mode control signals pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

(7) In some embodiments, the workholder includes a communications interface in communication with a control computer and in control communication with workholder hardware; and receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes includes receiving a set of mode control signals from the control computer to the communications interface.

(8) In some embodiments, the specified pre-defined mode includes a parameter defining a receiving width of a workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to open the workpiece interface to the receiving width.

(9) In some embodiments, the specified pre-defined mode includes a parameter defining a clamping width of a workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes

causing the workholder to close the workpiece interface to the clamping width.

(10) In some embodiments, the specified pre-defined mode includes a parameter defining a clamping force to be applied by a workpiece interface of the workholder to the workpiece; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to close the workpiece interface to apply the clamping force.

(11) In some embodiments, the specified pre-defined mode includes a parameter defining a vacuum pressure of a workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to apply the vacuum pressure to the workpiece.

(12) In some embodiments, the specified pre-defined mode includes a parameter defining a voltage applied to a workholder actuator of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to apply said voltage to the workholder actuator of the workholder.

(13) In some embodiments, the specified pre-defined mode includes a parameter defining a time duration to jiggle the workpiece interface upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to jiggle the workpiece for the specified time duration.

(14) In some embodiments, the specified pre-defined mode includes a parameter defining a vibration intensity at which to jiggle the workpiece interface upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to jiggle the workpiece at the specified vibration intensity.

(15) Another embodiment includes a workholder apparatus configured for sequentially holding each workpiece of a plurality of workpieces, each workpiece having a distinct holding specification from the respective holding specifications of other workpieces of the plurality of workpieces. In such embodiments, the workholder includes a workpiece interface controllable to open to receive the workpiece in an open configuration, and to close to grasp the workpiece in a closed configuration; and an actuator integral to the workholder and mechanically coupled to the workpiece interface. The workholder also includes a control circuit integral to the workholder, the control circuit configured to (1) receive specification of a pre-defined mode from a plurality of selectable pre-defined modes, each such mode specifying a plurality of holding parameters corresponding to the holding specification of a workpiece of the plurality of workpieces, said pre-defined mode being a specified pre-defined mode, and to (2) autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode.

(16) In some such embodiments, the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes. In such embodiments, to receive, at the control circuit, specification of a pre-defined mode from the plurality of pre-defined modes includes receiving at the control circuit control signals pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

(17) In some embodiments, the workholder apparatus includes a communications interface and in control communication with workholder hardware. In such embodiments, to receive, at the control circuit, specification of a pre-defined mode from the plurality of pre-defined modes includes receiving a set of mode control signals from the control computer to the communications interface.

(18) In some embodiments, the specified pre-defined mode includes a parameter defining a clamping width of the workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to close the workpiece interface to the clamping width.

(19) In some embodiments, the specified pre-defined mode includes a parameter defining a time duration to jiggle the workpiece interface and a vibration intensity at which to jiggle the workpiece interface, upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode includes causing the workholder to jiggle the workpiece at the specified vibration intensity for the specified time duration.

(20) Another embodiment includes a non-transitory computer readable medium having non-transient computer-executable code, the non-transient computer-executable code for controlling a workholder for autonomously executing holding operations pursuant to parameters of a specified pre-defined mode. In such embodiments, the computer-executable code includes: code for causing the workholder to selectively execute workholding operations of each mode of a plurality of distinct, selectable pre-defined modes, each such mode specifying a plurality of holding parameters corresponding to the holding specification of a workpiece of a plurality of workpieces; code for receiving, at a control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes, said pre-defined mode being a specified pre-defined mode; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode.

(21) In some such embodiments, the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes. In such embodiments, code for receiving, at the control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes includes code for receiving specification of a pre-defined mode pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

(22) In some embodiments, the workholder includes a communications interface, and code for receiving, at the control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes includes code for receiving a set of mode control signals from a control computer at the communications interface.

(23) In some embodiments, the specified pre-defined mode includes a parameter defining a receiving width of a workpiece interface of the workholder; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode includes code for causing the workholder to open the workpiece interface to the receiving width.

(24) In some embodiments, the specified pre-defined mode includes a first parameter defining a receiving width of a workpiece interface of the workholder; and a second parameter defining a clamping width of a workpiece interface of the workholder. In such embodiments, code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode includes code for causing the workholder to open the workpiece interface to the receiving width; and code for receiving a closing trigger signal subsequent to opening the workpiece interface to the receiving width; and code for causing the workholder to close the workpiece interface to the clamping width in response to receipt of the closing trigger signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Those skilled in the art should more fully appreciate advantages of various embodiments from the following “Description of Illustrative Embodiments,” discussed with reference to the drawings summarized immediately below.

- (2) FIG. 1A schematically illustrates a coordinate measuring machine, a robot and a storage apparatus for storing workpieces;
- (3) FIG. 1B schematically illustrates an embodiment of a coordinate measuring machine;
- (4) FIG. 1C schematically illustrates an embodiment of a workpiece;
- (5) FIG. 1D an embodiment of a control system for a coordinate measuring machine;
- (6) FIG. 1E schematically illustrates an embodiment of a manual user interface for a coordinate measuring machine;
- (7) FIG. 2 schematically illustrates an embodiment of a storage apparatus for storing workpieces;
- (8) FIG. 3A schematically illustrates an embodiment of a workpiece placement robot;
- (9) FIG. 3B schematically illustrates an embodiment of a workpiece placement robot;
- (10) FIG. 3C schematically illustrates an embodiment of a workpiece placement robot;
- (11) FIG. 3D schematically illustrates an embodiment of a workpiece placement robot;
- (12) FIG. 4A schematically illustrates an embodiment of a workholder;
- (13) FIG. 4B schematically illustrates another embodiment of a workholder;
- (14) FIG. 4C schematically illustrates an embodiment of a workpiece interface;
- (15) FIG. 4D schematically illustrates an embodiment of a workpiece interface;
- (16) FIG. 4E schematically illustrates an embodiment of a workpiece interface;
- (17) FIG. 4F schematically illustrates and embodiment of a workpiece;
- (18) FIG. 5A is a flowchart of an embodiment of a method of sequentially measuring a set of workpieces using a workpiece inspection system;
- (19) FIG. 5B is a flowchart of an embodiment of a method of operating a workholder;
- (20) FIG. 6A schematically illustrates a ruleset;
- (21) FIG. 6B schematically illustrates correlations between workpieces and corresponding rulesets;
- (22) FIG. 7A schematically illustrates an embodiment of an autonomous workholder;
- (23) FIG. 7B schematically illustrates an embodiment of an autonomous workholder; and
- (24) FIG. 7C schematically illustrates an embodiment of an autonomous workholder in communication an embodiment of a controller;
- (25) FIG. 8A and FIG. 8B schematically illustrates an embodiment of a workholder **400** in an “egg” holding mode;
- (26) FIG. 8C and FIG. 8D schematically illustrates an embodiment of a workholder in an “brick” holding mode;
- (27) FIG. 9A is a flowchart of a method of operating an autonomous workholder;
- (28) FIG. 9B is a flowchart of an embodiment of execution of workholder operations of specified mode.

DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

- (29) Illustrative embodiments improve operation of an industrial process (such as measuring the workpiece with a coordinate measuring machine, or machining the workpiece by a machine tool) by employing an autonomous workholder, which reduces workpiece inspection time, thereby increasing the efficiency of a process of inspecting a plurality of workpieces, such as a plurality of non-identical workpieces, and reducing or eliminating the risk of human operator error in manipulating workpieces in association with such a process. In illustrative embodiments, an autonomous workholder has a plurality of distinct modes, each distinct mode correlated to an associated workpiece for family of workpieces, the autonomous workholder configured to autonomously execute each such distinct mode.
- (30) One illustrative embodiment includes a system for inspecting each workpiece of a plurality of non-identical workpieces, each workpiece having a distinct workholding specification. The system includes a workholder configured to autonomously execute a plurality of workholding operations pursuant to parameters of a specified pre-defined workholding mode specified from a plurality of distinct, pre-defined workholding modes. Each pre-defined workholding mode specifies a plurality of holding parameters corresponding to the holding specification of a corresponding workpiece.

(31) Some embodiments include placing the workpiece on or in a workpiece interface of a workholder and, prior to securing the workpiece on or in the workholder, vibrating the workpiece interface to settle the workpiece onto or into the workpiece interface. The act of vibrating the workpiece interface is separate and distinct from an act of securing the workpiece to the workpiece interface, and the vibration from the act of vibrating the workpiece interface is separate and distinct from vibration that may occur incidental to the act of securing the workpiece to the workpiece interface. Some embodiments of a workholder include a vibration actuator distinct from a workpiece interface actuator that opens and closes the workpiece interface. Some embodiments of workpiece interface include a set of one or more tapered guides to guide a workpiece onto the workpiece interface.

(32) Definitions: As used in this description and the accompanying claims, the following terms shall have the meanings indicated, unless the context otherwise requires.

(33) An “actuator” is an apparatus coupled to a workholder and configured to move controllably in response to a controller. Such controllable motion may include a vibration to cause the workholder to open and/or close, for example to clamp and/or release a workpiece. An actuator may be, without limitation, an electric motor, a pneumatic apparatus, or a hydraulic apparatus, to name but a few examples.

(34) The terms “autonomous” and “autonomously” mean that the workholder **400** is configured to execute workholding operations pursuant to parameters specified by a workholding mode without further input from a source external to the workholder **400**. Execution of workholding operations by a workholder **400** shall be considered to be autonomous, and a workholder **400** shall be considered to execute workholding operations autonomously, even if one or more steps of such execution, after at least one step has been initiated, shall be initiated by, or dependent on receipt of, a trigger to initiate such one or more steps.

(35) A “closing motion,” in reference to a motion of a workpiece interface, is a motion in a direction so as to close the workpiece interface. A closing motion, if maintained, will ultimately cause the workpiece interface to clamp a workpiece to the workholder of which the workpiece interface is a part. However, a closing motion may be stopped before causing the workpiece interface to clamp a workpiece to the workholder.

(36) The term “end effector” (or simply “effector”) is a general term for an apparatus disposed on or integral to a robot arm, which apparatus is configured to get and hold an object to enable the robot arm to pick-up an object at one location, move and deliver the object to a different location. For example, one embodiment of an end effector is a mechanism used to grasp and hold an object to or by a robotic arm, typically (but not necessarily) disposed at the end of the robotic arm. An illustrative embodiment of such a mechanism is a gripper with two or more fingers.

(37) A “family” of workpieces means a set of workpieces, wherein each workpiece of said set is associated with the same (or an identical) workpiece delivery ruleset for customizing the configuration and/or the operation of at least one instrument of the set of instruments of a workpiece inspection system to move a workpiece and deliver the workpiece to a workholder. Delivering a workpiece to a workholder may include the operations (workholder operations) of the workholder itself. The workpieces in said set of workpieces may be identical to one another, or may be non-identical to one another, as long as the customization or configuration of said set of instruments of a workpiece inspection system is performed pursuant to the same (or an identical) workpiece delivery ruleset.

(38) This allows a robot **300** and/or workholder **400** to be configured pursuant to one workpiece delivery ruleset, even when the workpieces that belong to the family are non-identical to one another. In other words, not every non-identical workpiece requires a corresponding non-identical ruleset.

(39) The term “non-identical” with regard to a plurality of workpieces means that the workpieces would not be identical to one another even if all such workpieces are devoid of manufacturing

defects or deviations. For example, a fan blade and a ball bearing would be non-identical to one another because they would not be identical even if each was devoid of manufacturing defects or deviations. A plurality of workpieces are considered to be identical to one another if they would be physically identical in the case that each workpiece exactly matched the same design specification, free of manufacturing defects. For example, two fan blades based on the same design specification may be considered to be identical to one another when they would be identical to one another but for manufacturing defects or deviations.

(40) An “opening motion,” in reference to a motion of a workpiece interface, is a motion in a direction so as to open the workpiece interface. An opening motion may be stopped before the workpiece interface is open to its maximum degree, and/or before the workpiece interface is open to a degree that a workpiece can be removed from the workpiece interface.

(41) A “set” includes at least one member. For example, and without limiting the generality of the definition, a set of workpieces includes at least one workpiece.

(42) The term “workpiece” means an object to be operated upon by an industrial machine. For example, a workpiece may be inspected by a workpiece inspection instrument, such as a coordinate measuring machine, and/or may be operated upon by a lathe or computer numerically controlled cutting machine, to name but a few examples.

(43) A “workholder” is an apparatus that couples to a workpiece to hold the workpiece stationary, for example when the workpiece is on a table of a coordinate measurement machine. The term workholder may include a clamp; a vise; pneumatic vice; a vacuum suction device; a chuck; and a three-jaw chuck, to name but a few examples.

(44) The term “workholding operations” means operations performed by a workholder **400**.

Workholding operations include, for example, one or more of opening a workpiece interface (for example, to an unclamping width); closing a workpiece interface (for example, to a clamping width; and/or to a clamping force and/or with a vacuum force); jiggling (or vibrating) a workpiece interface; delaying an operation for a specified delay time, for example as measured from a trigger signal or from completion of a preceding operation.

(45) The term “holding specification” (or “workholding specification”) with regard to a workpiece means a set of parameters that define operation of a workholder for acquiring the workpiece, and/or holding the workpiece while the workpiece is being inspected by an inspection instrument, and/or releasing the workpiece. A holding specification for a workpiece may include, for example, a width to which a workholder opens to receive the workpiece; and/or the width to which the workholder closes to grasp the workpiece; and/or the speed at which the workpiece interface closes to grasp the workpiece; and/or the pressure applied by the workpiece interface to the workpiece, and/or jiggling (or vibrating) the workpiece interface (e.g., the intensity and/or duration of jiggling); and/or delaying an operation of the workholder, to name but a few examples. In embodiments in which a workholder secures a workpiece by subjecting the workpiece to a vacuum force, holding specification for a workpiece may include, for example, the pressure of a vacuum by which the workholder holds the workpiece, and/or jiggling (or vibrating) the workpiece interface (e.g., the intensity and/or duration of jiggling); and/or delaying an operation of the workholder, to name but a few examples. The “intensity” of jiggling (or vibrating) the workpiece interface is specified, in some embodiments, as the frequency at which the workpiece interface repeats cycles of alternating opening motions and closing motions, and in some embodiments the “intensity” of jiggling (or vibrating) the workpiece interface is specified as the number of open/close cycles per unit time of the workpiece interface. The “intensity” of jiggling (or vibrating) the workpiece interface is specified, in some embodiments, as the frequency at which a vibration actuator (**430**) rotates or vibrates.

(46) To “measure” an object means to determine a quantitative physical dimension of a portion of an object from a plurality of points on the object. A quantitative physical dimension may include, for example and without limitation, a height, length and/or width of the object or a portion of the

object. Measuring an object is not the same as merely detecting the presence of an object, and is not the same as merely locating an object, or determining the orientation of an object.

(47) To “detect” an object means to determine the presence of an object. It is possible to detect an object without measuring the object, or a portion of the object, in that detection does not require determining a quantitative physical dimension of a portion of an object.

(48) To “locate” an object means to determine the location of an object with reference to a known point in space, or within a coordinate system. It is possible to locate an object without measuring the object, or a portion of the object, in that locating an object does not require determining a quantitative physical dimension of a portion of an object.

(49) Environment

(50) FIG. 1A schematically illustrates a working environment for various embodiments. As shown the environment includes several instruments which may be referred to collectively as an embodiment of a workpiece inspection system **90**, including in this embodiment a coordinate measuring machine **100**, and a storage apparatus **200**, and a robot **300**. Some embodiments also include a workholder **400**, as described below.

(51) Coordinate Measuring Machine **100**

(52) As known by those in the art, a coordinate measuring machine (or “CMM”) **100** is a system configured to measure one or more features of a workpiece. Coordinate measuring machines are represented in FIG. 1A by coordinate measuring machine **100**.

(53) FIGS. 1B-1E schematically illustrate a coordinate measurement machine **100** that may be configured in accordance with illustrative embodiments.

(54) As known by those in the art, a CMM is a system configured to measure one or more features of a workpiece **180**. An illustrative embodiment of a workpiece **180** is schematically illustrated in FIG. 1C. Typically, a workpiece **180** has a specified shape with specified dimensions, which may be referred-to collectively as the “geometry” **181** of the workpiece **180**. As an example, a workpiece **180** may have an edge **182**, and a corner **183**. A workpiece **180** may also have surfaces, such as a flat surface **184**, and a curved surface **185**. A meeting of two surfaces may create an inside angle **187**. Moreover, each surface may have physical characteristic such as waviness **188** and/or surface finish **189**, as known in the art. A workpiece **180** may also have a cavity **186**, which may also be an aperture through the workpiece **180**. As known in the art, a cavity **186** may have dimensions such as width and depth, which may in turn define an aspect ratio of the cavity **186**.

(55) CMM Base

(56) In the illustrative embodiment of FIG. 1A, the CMM **100** includes a base **110** having a table **111**. The table **111** of the CMM **100** defines an X-Y plane **112** that typically is parallel to the plane of the floor **101**, and a Z-axis normal to the X-Y plane, and a corresponding X-Z plane and Y-Z plane. The table **111** also defines a boundary of a measuring space **113** above the table **111**. In some embodiments, the CMM **100** includes a probe rack **115** configured to hold one or more measuring sensors **140**. A moveable part of the CMM **100** may move to the probe rack **115** and place a measuring sensor **140** into the probe rack **115**, and/or remove another measuring sensor **140** from the probe rack **115**.

(57) Moveable Parts

(58) The CMM **100** also has movable features (collectively, **120**) arranged to move and orient a measuring sensor **140** (and in some embodiments, a plurality of such devices) relative to the workpiece **180**. As described below, movable features of the CMM **100** are configured to move and orient the measuring sensor **140**, relative to the workpiece **180**, in one dimension (X-axis; Y-axis; or Z-axis), two dimensions (X-Y plane; X-Z plane; or X-Y-Z plane), or three dimensions (a volume defined by the X-axis, Y-axis, and Z-axis). Accordingly, the CMM **100** is configured to measure the location of one or more features of the workpiece **180**.

(59) The CMM **100** of FIG. 1B is known as a “bridge” CMM. Movable features **120** of the bridge CMM **100** include a bridge **123** movably coupled to the base **110** by legs **121**. The bridge **123** and

legs **121** are controllably movable relative to the base **110** along the Y-axis.

(60) To facilitate motion of the legs relative to the base **110**, the legs **121** may be coupled to the base **110** by one or bearings **128**. As known in the art, a bearing may be a roller bearing or an air bearing, to name but a few examples.

(61) The movable features also include a carriage **125** movably coupled to the bridge **123**. The carriage is configured to controllably move in the X-axis along the bridge **123**. The position of the carriage **125** along the bridge **123** may be determined by a bridge scale **124** operably coupled to the bridge **123**.

(62) A spindle **126** is moveably coupled to the carriage **125**. The spindle **126** is configured to controllably move in the Z-axis. The position in the Z-axis of the spindle **126** may be determined by a spindle scale **127** operably coupled to the spindle **126**. The measuring sensor **140** is operably coupled to the spindle **126**. Consequently, the measuring sensor **140** is controllably movable in three dimensions relative to a workpiece **180** in the measuring space **113**.

(63) In some embodiments, the measuring sensor **140** is moveably coupled to the spindle **126** by an articulated arm **130**. For example, the measuring sensor **140** may be movably coupled to the arm **130** by a movable joint **131**. The moveable joint **131** allows the orientation of the measuring sensor **140** to be controllably adjusted relative to the arm **130**, to provide to the measuring sensor **140** additional degrees of freedom in the X-axis, Y-axis, and/or Z-axis.

(64) In other embodiments, which may be generally referred-to as “gantry” CMMs, the legs **121** stand on the floor **101**, and the measuring space **113** is defined relative to the floor **101**.

(65) In yet other embodiments, the measuring sensor **140** is fixed to (i.e., not movable relative to) the base **110**, and the table **111** is movable in one, two or three dimensions relative to the measuring sensor **140**. In some coordinate measuring machines, the table **111** may also be rotatable in the X-Y plane. In such embodiments, the CMM **100** moves the workpiece **180** relative to the measuring sensor.

(66) In other embodiments, which may be generally referred-to as “horizontal arm” CMMs, the bridge **123** is movably coupled to the base **110** to extend in the Z-axis, and to be controllably movable along the Y-axis. In such a CMM, the arm **130** is controllably extendable in the Z-axis, and controllably movable up and down the bridge **123** in the Z-axis.

(67) In yet other embodiments, the arm **130** is articulated. One end of the arm **130** is fixed to the base **110**, and a distal end of the arm **130** is movable relative to the base **110** in one, two or three dimensions relative to a workpiece **180** in the measuring space **113**.

(68) Sensors

(69) In some embodiments, the measuring sensor **140** may be a tactile probe (configured to detect the location of a point on the workpiece **180** by contacting a probe tip to the workpiece **180**, as known in the art), a non-contact probe (configured to detect the location of a point on the workpiece **180** without physically contacting the workpiece **180**), such as a capacitive probe or an inductive probe as known in the art, or an optical probe (configured to optically detect the location of a point on the workpiece **180**), to name but a few examples.

(70) In some embodiments, the measuring sensor **140** is a vision sensor that “sees” the workpiece **180**. Such a vision sensor may be a camera capable of focusing on the workpiece **180**, or the measurement space **113**, and configured to capture and record still images or video images. Such images, and/or pixels within such images, may be analyzed to locate the workpiece **180**; determine the placement and/or orientation of the workpiece **180**; identify the workpiece **180**; and/or measure the workpiece **180**, to name but a few examples.

(71) Some embodiments of a CMM **100** may include one, or more than one, camera **141** configured such that the measurement space **113** is within the field of view of the camera **141**. Such a camera **141** may be in addition to a measuring sensor **140**. The camera **141** may be a digital camera configured to capture still images and/or video images of the measurement envelope **113**, a workpiece **180** on the CMM **100**, and/or the environment around the CMM **100**. Such images may

be color images, black and white images, and/or grayscale image, and the camera **141** may output such images as digital data, discrete pixels, or in analog form.

(72) Some embodiments of a CMM **100** may also include an environmental sensor **142** configured to measure one or more characteristics of the environment **102** in which the CMM is placed, and some embodiments may have more than one such environmental sensor **142**. For example, an environmental sensor **142** may be configured to measure the temperature, pressure, or chemical content of the environment **102** around the CMM **100**. An environmental sensor **142** may also be a motion sensor, such as an accelerometer or a gyroscope, configured to measure vibrations of the CMM caused, for example, the by motion of people or objects near the CMM **100**. An environmental sensor **142** may also be a light detector configured to measure ambient light in the environment **102**, which ambient light might, for example, interfere with the operation of an optical sensor or vision sensor. In yet another embodiment, an environmental sensor **142** may be sound sensor, such as a microphone, configured to detect sound energy in the environment.

(73) In operation, the CMM **100** measures the workpiece **180** by moving the measuring sensor **140** relative to the workpiece **180** to measure the workpiece **180**.

(74) CMM Control System

(75) Some embodiments of a CMM **100** include a control system **150** (or “controller” or “control logic”) configured to control the CMM **100**, and process data acquired by the CMM. FIG. 1D schematically illustrates an embodiment of a control system **150** having several modules in electronic communication over a bus **151**.

(76) In general, some or all of the modules may be implemented in one or more integrated circuits, such as an ASIC, a gate array, a microcontroller, or a custom circuit, and at least some of the modules may be implemented in non-transient computer-implemented code capable of being executed on a computer processor **157**.

(77) Some embodiments include a computer processor **157**, which may be a microprocessor as available from Intel Corporation, or an implementation of a processor core, such as an ARM core, to name but a few examples. The computer processor **157** may have on-board, digital memory (e.g., RAM or non-transient ROM) for storing data and/or computer code, including non-transient instructions for implementing some or all of the control system operations and methods.

Alternately, or in addition, the computer processor **157** may be operably coupled to other digital memory, such as RAM or non-transient ROM, or a programmable non-transient memory circuit for storing such computer code and/or control data. Consequently, some or all of the functions of the controller **150** may be implemented in software configured to execute on the computer processor.

(78) The control system **150** includes a communications interface **152** configured to communicate with other parts of the CMM **100**, or with external devices, such as computer **170** via communications link **176**. To that end, communications interface **152** may include various communications interfaces, such as an Ethernet connection, a USB port, or a Firewire port, to name but a few examples.

(79) The control system **150** also includes a sensor input **155** operably coupled to one or more sensors, such as a measuring sensor **140** or camera **141**. The sensor input **155** is configured to receive electronic signals from sensors, and in some embodiments to digitize such signals, using a digital to analog (“D/A”) converter. The sensor input **155** is coupled to other modules of the control system **150** to provide to such other modules the (digitized) signals received from sensors.

(80) The motion controller **153** is configured to cause motion of one or more of the movable features **120** of the CMM **100**. For example, under control of the computer processor **157**, the motion controller **153** may send electrical control signals to one or more motors within the CMM **100** to cause movable features of the CMM **100** to move a measuring sensor **140** to various points within the measuring space **113** and take measurements of the workpiece **180** at such points. The motion controller **153** may control such motion in response to a measurement program stored in memory module **156**, or stored in computer **170**, or in response to manual control by an operator

using manual controller **160**, to name but a few examples.

(81) Measurements taken by the CMM **100** may be stored in a memory module **156**, which includes a non-transient memory. The memory module **156** is also configured to store, for example, a specification for a workpiece **180** to be measured; a specification for a calibration artifact; an error map; and non-transient instructions executable on the computer processor **157**, to name but a few examples. Such instructions may include, among other things, instructions for controlling the moveable features of the CMM **100** for measuring a workpiece **180** and/or a calibration artifact; instructions for analyzing measurement data; and instructions for correcting measurement data (e.g., with an error map).

(82) The measurement analyzer **154** is configured to process measurement data received from one or more sensors, such as measuring sensor **140**. In some embodiments, the measurement analyzer **154** may revise the measurement data, for example by modifying the measurement data using an error map, and/or compare the measurement data to a specification, for example to assess deviation between a workpiece **180** and a specification for that workpiece **180**. To that end, the measurement analyzer **154** may be a programmed digital signal processor integrated circuit, as known in the art.

(83) Alternately, or in addition, some embodiments couple the CMM **100** with an external computer (or “host computer”) **170**. In a manner similar to the control system **150**, the host computer **170** has a computer processor such as those described above, and non-transient computer memory **174**, in communication with the processor of the CMM **100**. The memory **174** is configured to hold non-transient computer instructions capable of being executed by the processor, and/or to store non-transient data, such as data acquired as a result of the measurements of an object **180** on the base **110**.

(84) Among other things, the host computer **170** may be a desktop computer, a tower computer, or a laptop computer, such as those available from Dell Inc., or even a tablet computer, such as the iPad™ available from Apple Inc. In addition to the computer memory **174**, the host computer **170** may include a memory interface **175**, such as a USB port or slot for a memory card configured to couple with a non-transient computer readable medium and enable transfer of computer code or data, etc. between the computer **170** and the computer readable medium.

(85) The communication link **176** between the CMM **100** and the host computer **170** may be a hardwired connection, such as an Ethernet cable, or a wireless link, such as a Bluetooth link or a Wi-Fi link. The host computer **170** may, for example, include software to control the CMM **100** during use or calibration, and/or may include software configured to process data acquired during operation of the CMM **100**. In addition, the host computer **170** may include a user interface configured to allow a user to manually operate the CMM **100**. In some embodiments, the CMM and/or the host computer **170** may be coupled to one or more other computers, such as server **179**, via a network **178**. The network **178** may be a local area network, or the Internet, to name but two examples.

(86) Because their relative positions are determined by the action of the movable features of the CMM **100**, the CMM **100** may be considered as having knowledge of the relative locations of the base **110**, and the workpiece **180**. More particularly, the computer processor **157** and/or computer **170** control and store information about the motions of the movable features. Alternately, or in addition, the movable features of some embodiments include sensors that sense the locations of the table **111** and/or measuring sensor **140**, and report that data to the computer **170** or controller **150**. The information about the motion and positions of the table and/or measuring sensor **140** of the CMM **100** may be recorded in terms of a one-dimensional (e.g., X, Y or Z), two-dimensional (e.g., X-Y; X-Z; Y-Z) or three-dimensional (X=Y-Z) coordinate system referenced to a point on the CMM **100**.

(87) Manual User Interface

(88) Some CMMs also include a manual user interface **160**. As shown in FIG. 1E, the manual user interface **160** may have controls (e.g., buttons; knobs, etc.) that allow a user to manually operate

the CMM **100**. Among other things, the interface **160** may include controls that enable the user to change the position of the measuring sensor **140** relative to the workpiece **180**. For example, a user can move the measuring sensor **140** in the X-axis using controls **161**, in the Y-axis using controls **162**, and/or in the Z-axis using controls **163**.

(89) If the measuring sensor **140** is a vision sensor, or if the CMM **141** includes a camera **141**, then the user can manually move the sensor **140**, camera **141**, or change field of view of the vision sensor and/or camera using controls **165**. The user may also focus the vision sensor and/or camera **141** using control **166** (which may be a turnable knob in some embodiments) and capture and image, or control recording of video, using control **167**.

(90) As such, the movable features may respond to manual control, or be under control of the computer processor **157**, to move the base **110** and/or the measuring sensor **140** relative to one another. Accordingly, this arrangement permits the object being measured to be presented to the measuring sensor **140** from a variety of angles, and in a variety of positions.

(91) Embodiments of a CMM **100** include a mobile controller which may be referred-to as a jogbox (or “pendant”) **190**. The jogbox **190** includes a number of features that facilitate an operator's control of the coordinate measuring machine **100**.

(92) The jogbox **190** is not affixed to the coordinate measuring machine **100** in that its location is movable relative to the coordinate measuring machine **100**. The mobility of the jogbox **190** allows an operator of the coordinate measuring machine **100** to move relative to the coordinate measuring machine **100**, and relative to a workpiece **180** on which the coordinate measuring machine **100** operates. Such mobility may allow the operator to move away from the coordinate measuring machine **100** for safety reasons, or to get a broader view of the coordinate measuring machine **100** or the workpiece **180**. The mobility of the jogbox **190** also allows the operator to move closer to the coordinate measuring machine **100** and the workpiece **180** on which it operates than would be possible using a fixed control console or computer **170**, in order, for example, to examine or adjust the location or orientation of the workpiece **180**, or the operation of the coordinate measuring machine **100**.

(93) To that end, the jogbox **190** is in data communication with the control system **150**, and may be movably coupled to the control system **150** by a tether **191**. In some embodiments, the jogbox **190** is in data communication with the communications interface **152** of the control system **150** via a tether **191** (which may be an Ethernet cable, a USB cable, or a Firewire cable, to name but a few examples), as schematically illustrated in FIG. 1B, and in other embodiments the jogbox **190** is in data communication with the communications interface **152** of the control system **150** via a wireless communications link, such as a Bluetooth connection, etc.

(94) Storage Apparatus **200**

(95) One or more workpieces **180** are stored in storage apparatus (or system) **200**, an embodiment of which is schematically illustrated in FIG. 2. In this embodiment, the storage system **200** includes one or more drawers or shelves **201**. The storage system defines a storage system coordinate system having three mutually orthogonal axes (axes X, Y and Z in FIG. 1A).

(96) As schematically illustrated in FIG. 1A, each drawer or shelf **211** of a storage system **200** may have one or more storage plates **203** configured and disposed to hold the one or more workpieces **180**. A storage plate **203** may have a plate surface **202**.

(97) Robot **300**

(98) A robot **300** is schematically illustrated in FIG. 1A, FIG. 3A relative to the three mutually orthogonal axes (X, Y and Z in FIG. 1A).

(99) In illustrative embodiments, robot **300** is disposed so that it can reach the drawer or shelf **201** of a storage apparatus **200**, and each workpiece **180** of a set of workpieces disposed at the storage apparatus **200**, as well as the measurement space **113** (e.g., table **111**) of the coordinate measuring machine **100**, and a set of workpieces on the storage apparatus **200** and coordinate measuring machine **100**. When disposed in that manner, the robot **300** can transport a workpiece **180** from the

drawer or shelf **201** to the measuring space **113** of the coordinate measuring machine **100**, and can transport a workpiece **180** from the measuring space **113** of the coordinate measuring machine **100** to the drawer or shelf **201**. To that end, the robot **300** in this embodiment has an effector **340**, typically at the end **303** of a movable, articulated arm **302**. In this embodiment, the end effector **340** is a gripper **311** at the end **303** of a movable, articulated arm **302**.

(100) In some embodiments, the gripper **311** has two or more fingers **314**, **315** separated by a gripper gap **317**. The gripper **311** is configured to controllably close and open the fingers **314**, **315** to decrease or increase the gripper gap **317** (respectively) so as to grasp and release (respectively) a workpiece **180**.

(101) In illustrative embodiments, the robot **300** (e.g., motion of the robot arm **302** and/or motion of the gripper **311**) is controlled by a robot controller. For example, in some embodiments, the robot **300** is controlled by robot control computer **379**, or a robot control interface **390**. In alternate embodiments, the robot **300** is controlled by the motion controller **153** or the host computer **170** of the coordinate measuring machine **100**, which are separate and distinct from the robot control computer **379** and the robot control interface **390**.

(102) In illustrative embodiments, the robot arm **302** includes sensors configured to measure the location of the end **303** of the arm **302** relative to the base **301** of the robot **300**, each location defined by a corresponding robot arm position datum.

(103) FIG. 3B, FIG. 3C, and FIG. 3D each schematically illustrates an alternate embodiment of a robot **300**, each of which is able to obtain a workpiece, move the workpiece, and deliver the workpiece to the measurement volume of a coordinate measuring machine **100** or other inspection instrument. The robot **300** in FIG. 3C has an arm **302** that is slidably coupled to base **301**. In operation, the arm **302** slides along the base **301**, in the X-axis, to move a workpiece in held by its effector **311**. The arm **302** may also move the effector **311**, and the workpiece, independently in the Y-axis and/or the Z-axis. The robot **300** in FIG. 3D has an arm **302** that is slidably and/or pivotably coupled to base **301**. In the operation of some embodiments, the arm **302** slides relative to the base **301** in the X-axis to move a workpiece held by its effector **311**, and/or pivots relative to the bases **301** to move the effector **311** and workpiece in the X-Y plane. The arm **302** may also move the effector **311**, and the workpiece, independently in the Y-axis and/or the Z-axis.

(104) FIG. 4A schematically illustrates an embodiment of a workholder **400** (which may also be referred-to as a workpiece “fixture”).

(105) The workholder **400** has a base **410**, which is configured to rest in a stable position on a surface, such as the table **111** of a coordinate measuring machine **100**, for example. In some embodiments, the workholder is affixed to the coordinate measuring machine **100**, and in some embodiments, the workholder **400** simply rests on the table **111** of the coordinate measuring machine **100**.

(106) The workholder **400** also has a workpiece interface **420** for receiving, securely holding, and releasing a workpiece **180**, for example in the process of inspecting the workpieces with an inspecting machine **100**. To that end, in this embodiment, the workpiece interface **420** has two clamp arms or jaws **421** and **422** (such as workpiece interface **420** may be referred to as a “clamp apparatus”). The jaws define a controllable workholder gap **425** between them, and the workpiece interface **420** may be referred-to as a clamp apparatus. For example, in some embodiments, both jaws **421** and **422** are movable relative to the base **410**, and in some embodiments only one of the jaws, **421** or **422**, is movable relative to the base. The workholder gap **425**, which is the distance between the jaws **421**, **422**, is automatically controllable and can be opened (i.e., the workholder gap **425** increased) or closed (the workholder gap **425** decreased). Moreover, when a workpiece **180** is disposed within the workpiece interface (e.g., clamped by the jaws **421**, **422**), the amount of force or pressure exerted on the workpiece **180** by the workholder **400** (e.g., by the jaws **421**, **422**) is controllable based on the specific workpiece or type of workpiece **180** being held by the workholder **400**. For example, a delicate workpiece **180** may be held with less clamping force

imposed on the workpiece **180** by the jaws **421, 422** than the force imposed by the jaws **421, 422** on a more robust workpiece **180**. In preferred embodiments, the clamping force imposed on the workpiece **180** by the jaws **421, 422** is sufficient to hold the workpiece **180** in a fixed position, relative to the workholder base **410**, during inspection by an inspection machine **100** (e.g., a coordinate measuring machine), so the inspection operations do not cause the workpiece **180** to move, wiggle, or shift positions in response to said inspection operations.

(107) Illustrative embodiments of a workholder **400** include, as an integral part of the workholder **400**, a computer processor **411**. The computer processor **411** may include a microprocessor from Intel or AMD, or a microprocessor based on an ARM core, or a microcontroller, to name but a few examples. The computer processor **411** may include a memory to store executable instructions (or “computer code”), which memory is accessible by the microprocessor or controller.

(108) The computer processor **411** is in control communication with a workholder actuator **413**, which is in control communication with one or more of the jaws **421, 422**. For example, in some embodiments one or more of the jaws **421, 422** is threadedly coupled to a threaded rod such that turning of the threaded rod by the workholder actuator **413** causes one or more of the jaws **421, 422** to move relative to one another. In such embodiments, the workholder actuator **413** causes the threaded rod to turn in a first direction to cause the jaws **421, 422** to execute a closing motion, and causes the threaded rod to turn in a second direction, the second direction being opposite to the first direction, to cause the jaws **421, 422** to execute an opening motion.

(109) In other embodiments, one or more of the jaws **421, 422** is coupled to a driving wheel via one or more corresponding connecting rods, such that turning of the driving wheel by the workholder actuator **413** causes one or more of such connecting rods to cause one or more of the jaws **421, 422** to move relative to one another.

(110) The computer processor **411** is configured to control the actuator **413** to customize the configuration of the workpiece interface **420**, for example to controllably open and close the workpiece interface gap **425** by moving one or more of the jaws **421, 422** pursuant to execution of computer code.

(111) Alternatively, or in addition to the computer processor **411**, some embodiments include a communications interface **415**. In some embodiments, the communications interface **415** is coupled in electronic communication with the workholder actuator **413**, and with a controller (or “control computer”) such as controller **91**, host computer **170**, or CMM controller **150**. Control signals from the controller cause the actuator **413** to customize the configuration of the workpiece interface **420** for example to controllably open and close the workpiece interface gap **425** by moving one or more of the jaws **421, 422** pursuant to control signals received at the workholder **400** via the communications interface **415**. In some embodiments, the communication interface **415** is coupled in data communication with computer processor **411** to instruct, control or operate the computer processor **411** to control the workholder actuator **413** to customize the configuration of the workpiece interface **420** for example to controllably open and close the workpiece interface gap **425** by moving one or both of the jaws **421, 422** pursuant to control signals received at the workholder **400** via the communications interface **415**.

(112) Some embodiments of a workholder **400** include a vibration actuator **430**. FIG. **4B** schematically illustrates an embodiment of such a workholder **400**. When activated, the vibration actuator **430** causes a physical vibration, which physical vibration causes the workholder **400**, or at least the workpiece interface **420**, to vibrate (or “jiggle”) with a force, intensity, and frequency sufficient to cause a workpiece **180** to settle onto and/or into the workpiece interface **420**. To that end, in illustrative embodiments, the vibration actuator **430** is physically coupled to the workholder **400**, and in some embodiments is directly physically coupled to the workpiece interface **420**, such that vibrations produced by the vibration actuator **430** cause physical vibration of the workpiece interface **420**.

(113) Some embodiments of the workholder **400** include a computer processor **411** as described

above. In some embodiments, however, the computer processor **411** is also in control communication with the vibration actuator **430**.

(114) Alternatively, or in addition to the computer processor **411**, some embodiments include a communications interface **415**. In some embodiments, the communications interface **415** is coupled in electronic communication with the vibration actuator **430**, to control the vibration actuator **430** to cause the workholder **400**, or the workpiece interface **420**, to vibrate (or “jiggle”). In some embodiments, the communications interface **415** is coupled in electronic communication with the computer processor **411** to provide executable code from an external memory to the computer processor **411**, and/or to provide control signals to the computer processor **411**.

(115) In some embodiments, the vibration actuator **430** includes an electric motor that vibrates or physically oscillates when activated. For example, such an electric motor may have a motor shaft that rotates about an axis, and that motor shaft includes a weight coupled to the motor shaft such that the weight rotates eccentrically when the motor shaft rotates about its axis, such that the eccentric motion causes vibration of the motor and, because the motor is coupled to the workholder **400** and/or to the workpiece interface, vibration of the workpiece interface **420**.

(116) In some embodiments, the vibration actuator **430** includes a fluid-driven apparatus, such as a pneumatic actuator. In such embodiments, the workholder may include a fluid connector **416** configured to receive a fluid to controllably drive the fluid-driven apparatus. Alternating activation of the fluid-driven apparatus causes the fluid-driven apparatus to vibrate.

(117) Embodiments of various workpiece interfaces **420** are schematically illustrated in FIG. 4C, FIG. 4D and FIG. 4E. Each workpiece interface **420** of FIG. 4C, FIG. 4D and FIG. 4E may each be referred to as a workpiece clamp in that the workpiece interface **420** is configured to close to secure a workpiece **180**, to hold the workpiece **180** during operations on the workpieces **180**, such as operations by a machine tool or a coordinate measuring machine, and to controllably release the workpiece **180**.

(118) FIG. 4C schematically illustrates an embodiment of a workpiece interface **420**. At least one clamp **421**, **422** of the workpiece interface **420**, and in some embodiments each clamp **421**, **422** of the workpiece interface **420**, includes a substrate **425**, and defines a workpiece interface plane **440** (which may also be referred-to as a workpiece plane **440**, or a substrate plane **440**), which in FIG. 4C is in the X-Y plane, or is parallel to the X-Y plane. In some embodiments, the workpiece interface plane **440** is coplanar with a surface of the substrate **425**, and in some embodiments the workpiece interface plane **440** is not coplanar with a surface of the substrate **425**. In use of the workpiece interface **420**, a workpiece **180** may rest on the substrate **425**.

(119) The workpiece interface **420** also includes a set of tapered guides, where the set includes at least one tapered guide, and in some embodiments the set includes more than one tapered guide.

(120) In some embodiments, a tapered guide includes a post **450** extending above the workpiece plane **440** in a direction away from the workpiece plane **440**. The post **450** includes a slide surface **451** angled to intersect the substrate plane **440** at an intersection **452** such that the slide surface **451** and the substrate plane **440** form an obtuse angle at said intersection **452**, the slide surface **451** disposed and configured to guide the workpiece **180** onto the substrate in response to vibrating the workpiece interface. In some embodiments, the slide surface **451** defines a slide surface plane, and is configured such that the plane intersects the substrate plane at an obtuse angle, such that the slide surface **451** is disposed and configured to guide the workpiece **180** onto the substrate in response to vibrating the workpiece interface.

(121) In some embodiments, the tapered post **450** is configured to pass through an aperture **480** in the workpiece **180** (which may be referred-to as a “workpiece aperture”), which workpiece aperture **480** has an edge surface **481**. Such a workpiece **180** is schematically illustrated in FIG. 4F. In such embodiments, the slide surface **451** is disposed and configured to engage the edge surface **481** to guide the workpiece **180** onto the substrate **425** in response to vibrating the workpiece interface **420**. For example, the edge surface **481** slides down the slide surface **451** in response to vibrating

the workpiece interface **420**. Taking the workpiece interface **420** of FIG. **4D** for example, one or more edge surfaces **481** engage one or more corresponding slide surfaces **451** to guide the workpiece **180** down onto or into the workpiece interface **420**, including causing the workpiece **180** to move in the X-axis (specifically, in the negative direction of the X axis illustrated in FIG. **4D**) as the edge surface **481** slides down the slide surface **451**.

(122) Some embodiments include a set of tapered guides, and each tapered guide has a surface extending away from the substrate plane, and disposed and configured to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

(123) Some embodiments include a corner guide **460** having two surfaces **461**, **462**, the two surfaces **461**, **462**, each extending upward away from the workpiece plane **440**. In some embodiments, the two surfaces **461**, **462** extend away from the workpiece plane **440** at a right angle (90 degrees, or orthogonal) to the workpiece plane **440**.

(124) The two surfaces **461**, **462** are disposed at an angle to one another, so as to guide the workpiece onto the substrate **425** in two corresponding directions in response to vibrating the workpiece interface. In FIG. **4C** the two corresponding directions are the X-axis for surface **462**, and the Y-axis for surface **461**.

(125) In some embodiments, one or both of the surfaces **461**, **462** are not tapered, in that they form a right angle with the workpiece plane **440**. In such embodiments, the surfaces **461**, **462** are configured to apply a clamping force to a workpiece **180**, to assist in securing the workpiece **180** to the workpiece interface **420**.

(126) In some embodiments, one or both of the surfaces **461**, **462** are tapered to form an acute angle with the workpiece plane **440** so as to urge a workpiece **180** down onto the substrate **425**. In other embodiments, one or each of the two surfaces **461**, **462** extend away from the workpiece plane **440** at an angle less than a right angle, such that the two surfaces **461**, **462** are at an obtuse angle relative to the workpiece plane **440**. Embodiments in which at least one of the surfaces **461**, **462** is tapered may be considered to be a tapered guide.

(127) Some embodiments include a pair of corner guides **460** disposed such that the pair of corner guides cooperate, in response to vibrating the workpiece interface, to automatically center the workpiece in at least one axis in the substrate plane **440**. In the embodiment of FIG. **4C**, a pair of corner guides, each numbered **460**, cooperate to center a workpiece **180** in the Y-axis, for example by holding the workpiece between the two corner guides **460**.

(128) Some embodiments of the workpiece interface **420** include a set of one or more end stops **470**. Each end stop **470** includes a wall **471** perpendicular to the substrate plane **440**. In illustrative embodiments, the wall **471** faces the direction in which the workpiece interface **420** closes. In illustrative embodiments, the wall **471** on one portion **421** of the workpiece interface **420** faces the other portion **422** of the workpiece interface **420**.

(129) In operation, the wall **471** engages the workpiece **180** to prevent the workpiece **180** from sliding off the workpiece interface **420**, for example when the workpiece interface **420** is in the process of closing. Moreover, each end stop **470** of the set of end stops is disposed and configured to constrain the workpiece **180** while the workpiece **180** is clamped by the workpiece interface **420**.

(130) Some embodiments of end stop **470** include a porch **472**. The port **472** maybe coplanar with the substrate plane **440**. In operation, the workpiece **180** rests on the porch **472**. In some embodiments, one or more tapered guides guide the workpiece **180** to rest on the porch **472** in response to jiggling of the workpiece interface **420**.

(131) Some embodiments include a pair of corner guides **460** and a set of one or more end stops **470**. Such embodiments constrain a workpiece **180** in three directions in (or parallel to) the substrate plane **440**. In the embodiment of FIG. **4C**, for example, the corner guides **460** and end stops **470** are configured to constrain a workpiece **180** resting on the substrate in the positive direction of the Y axis, and the negative direction of the Y axis, and in one direction of the X axis. The workpiece remains free to move in the other direction of the X axis.

(132) FIG. 5A is a flowchart of an embodiment of a method 500 of sequentially measuring a series of workpieces 180 using an inspection system 90. The inspection system 90 customizes the configuration of one or more apparatuses of the inspection system 90 (e.g., robot 300), and/or customizes the operation of one or more apparatuses of the inspection system 90 (e.g., robot 300; workholder 400), to meet the requirements of each workpiece, for example where the workpieces are from different families of workpieces. Parameters for adapting apparatuses and/or operations of apparatuses are stored in a ruleset 610 corresponding to each workpiece 180 (or corresponding to a family to which the workpiece 180 belongs), as described below, and are read by the controller 91, for example from a memory within or accessible by the controller 91, which controller then causes the adaption of the apparatuses and operations accordingly. In some embodiments, the memory within or ruleset database 92.

(133) The method 500 includes, at step 510, providing the inspection system 90. In some embodiments, providing the inspection system 90 includes providing a workpiece inspection machine (e.g., coordinate measuring machine 100), and/or a workpiece storage apparatus 200, and/or a robot 300, and/or a controller 91.

(134) The method 500 includes, at step 520, providing a plurality of workpieces for inspection by the inspection instrument. In some illustrative embodiments, the workpieces of the plurality of provided workpieces are non-identical to one another. In some illustrative embodiments, each workpiece of the plurality of provided workpieces belongs to a different family of a plurality of families of workpieces. In other words, each workpiece of the plurality of workpieces being from a different family of a plurality of families of workpieces. See, for example, workpieces 686 and 687 of family 684, and workpieces 688 and 689 of family 685, in FIG. 6B. Consequently, there are a plurality of families of workpieces, and each workpiece 180 may be said to “belong” to a corresponding one of the families of workpieces.

(135) The method 500 includes, at step 525, providing a plurality of rulesets. Each ruleset 610 of the plurality of rulesets corresponds, respectively, to a family of the plurality of families of workpieces, and may be described as a “corresponding” ruleset for said family. See, for example, ruleset 624 corresponding to the workpieces 686 and 687 of family 684, and ruleset 625 corresponding to the workpieces 688 and 689 of family 685, in FIG. 6B.

(136) Each corresponding ruleset includes parameters pursuant to which the controller 91 customizes a set of one or more instruments of the inspection system 90 to inspect a workpiece 180 from the family to which the workpiece 180 belongs. In illustrative embodiments, the plurality of rulesets are stored in a database in data communication with controller 91, or stored in a memory (e.g., a non-volatile memory) of controller 91.

(137) The method 500 includes, at step 530, obtaining a workpiece 180 to be inspected by a workpiece inspection instrument (e.g., coordinate measuring machine 100), and obtaining a ruleset (a “corresponding ruleset”) corresponding to that workpiece 180, such a ruleset corresponding to the family to which that workpiece belongs. In illustrative embodiments, the corresponding ruleset is retrieved, by the controller, from the database or memory in which a plurality of workpiece delivery rulesets is stored. In illustrative embodiments, step 530 includes retrieving, from the plurality of workpiece delivery rulesets, a ruleset corresponding to the family of said workpiece, said corresponding ruleset comprising a set of parameters to automatically customize transfer of the workpiece to a workholder.

(138) Historically, obtaining a workpiece 180 has been done by having an operator provide the workpiece 180, or by having an operator manipulate a robot 300 to obtain the workpiece 180.

(139) Some robots may be able to retrieve an object from a location automatically without operator intervention if the location of the object is accurately known to the robot, but in such cases conventional robots can only follow pre-programmed instructions, and lack the ability to adapt their actions to changing conditions. For example, conventional robots cannot automatically adapt their behavior to operate differently for different (e.g., non-identical) workpieces. Sometimes, when

consecutively obtaining two workpieces **180** which workpieces **180** are not identical to one another, the robot's operation for obtaining the first workpiece **180** may not be appropriate for obtaining the second workpiece **180**, such as when the second workpiece is more delicate than the first workpiece and therefore requires a lower gripping pressure by the gripper **311** than the first workpiece non-transient, or such as when the second workpiece **180** has a different shape than the first workpiece **180**, and therefore requires that the gripper **311** grasp the second workpiece **180** in a location on the second workpiece **180** that is specific to that second workpiece **180**, and which would not be possible or viable for grasping the first workpiece **180**.

(140) In illustrative embodiments, obtaining a workpiece **180** includes moving a robot arm **302** to the location of the workpiece **180** (e.g., storage **200**) and grasping the workpiece **180** with an effector (e.g., robot gripper **311**).

(141) The method **500** includes, at step **535**, customizing a set of instruments of the system. Step **535** may be described as customizing the transfer of workpieces to a workholder. In illustrative embodiments, one or more instruments of the set of instruments, and/or the operation one or more instruments of the set of instruments, are customized by the controller **91** pursuant to parameters from the corresponding ruleset for the particular workpiece **180** being moved. In other words, the controller **91** customizes (i) the set of instruments and/or (ii) the operation of the set of instruments pursuant to parameters from the corresponding ruleset. In some embodiments, step **535** includes sequentially, using the control system to: (a) customize at least one of (i) the configuration of the robot, and (ii) the operation of the robot, pursuant to the parameters of the corresponding ruleset; and subsequently (b) operate the robot to deliver said non-identical workpiece to the workholder.

(142) In illustrative embodiments, automatically grasping a workpiece **180** (e.g., when the workpiece **180** is at a storage apparatus **200**) by a robot **300** may involve one or more parameters (e.g., in a ruleset **610**) that define aspects of the grasping operation. In illustrative embodiments, each workpiece **180** has a set of parameters that are specific to that workpiece **180** (and workpieces that are identical to that workpiece **180**).

(143) For example, grasping a first workpiece **180** may require the gripper fingers **314**, **315** to be open to a gripper gap **317** of a first width prior to grasping the first workpiece **180**. Consequently, the gripper gap **317** width for the first workpiece **180** may be a parameter in a first robot ruleset, which first robot ruleset corresponds to the first workpiece **180**.

(144) However, that gripper gap **317** width may not be sufficient for a second workpiece **180**, for example if the second workpiece **180** requires the gripper fingers **314**, **315** to be open to a gripper gap **317** of a second width, which is greater than the first width, prior to grasping the second workpiece **180**. For example, the gripper **311** may need to open the gripper fingers **314**, **315** to a gap of only 2 centimeters to grasp the first workpiece **180**, but if the second workpiece has a diameter of 3 centimeters, then the gripper **311** may need to open the gripper fingers **314**, **315** to a gap of 3 or 4 centimeters to grasp the second workpiece **180**. Such adjustment and adaptations are easy for a human operator, but not conventionally automatically possible for a robot **300**.

Moreover, even a competent and experienced human operator can make a mistake and fail to make such an adjustment or adaptation, and may consequently damage the robot **300** and/or a workpiece **180**, such as by causing the gripper **311** to collide with the workpiece **180**, or by holding the workpiece **180** too loosely, allowing the workpiece **180** to shift positions within the gripper **311**, or fall out of the gripper **311** entirely, in either case incurring damage.

(145) Consequently, the gripper gap width **317** for the second workpiece **180** may be a parameter in a second robot ruleset, which second robot ruleset corresponds to the second workpiece **180**. In operation, the controller **91** will read the gripper gap parameter from the first robot ruleset and cause the robot **300** to open the gripper fingers **314**, **315** to the first gripper gap when obtaining the first workpiece. Similarly, the controller **91** will read the gripper gap parameter from the second robot ruleset and cause the robot **300** to open the gripper fingers **314**, **315** to the second gripper gap when obtaining the second workpiece.

(146) Similarly, for grasping workpieces **180**, grasping a first workpiece **180** for a first family of workpieces **180** may require the gripper fingers **314**, **315** to be closed to a first closed gripper gap **317** of a first width when closing the gripper fingers **314**, **315** around the first workpiece **180**. Consequently, the first robot ruleset may include a parameter specifying the gripper width **317** of the gripper fingers **314**, **315** when grasping the first workpiece **180** from the first family of workpieces, and the controller **91** will read that parameter and cause the robot **300** to close gripper fingers **314**, **315** accordingly to grasp the first workpiece **180**. Similarly, a second robot ruleset may include a parameter specifying the gripper width **317** of the gripper fingers **314**, **315** when grasping the second workpiece **180** of a different (e.g., second) family of workpieces, and the controller **91** will read that parameter and cause the robot **300** to close gripper fingers **314**, **315** accordingly to grasp the second workpiece **180** from the second family of workpieces. In illustrative embodiments, a gripper gap **317** parameter may be specified as a quantitative distance (e.g., 2 mm, 4 mm, etc.), or may be specified in terms of the maximum and/or minimum width of the gripper gap **317** (e.g., open to the minimum gripper gap **317**; close all the maximum gripper gap **317**; close to 50% of the maximum gripper gap **317**). In other embodiments, a gripper gap **317** parameter may be specified in terms of a force or pressure exerted by the gripper on a workpiece **180** (e.g., close the gripper gap **317** until the gripper exerts a specified quantitative pressure is on the workpiece; open the gripper gap **317** until force or pressure exerted on the workpiece **180** by the gripper is at (or is reduced to) a specified quantitative pressure).

(147) Next, the method includes operating the robot **300** to deliver said non-identical workpiece to the workholder **400**.

(148) To that end, the method **500** includes, at step **540**, moving the workpiece **180**, in the grasp of the gripper **311**, to the inspection instrument. For example, step **540** includes, in some embodiments, moving the workpiece **180** to the measurement envelope **113** of the coordinate measurement machine **100**. In illustrative embodiments, this includes moving the robot arm **302** so that the workpiece **180**, in the grasp of the gripper **311**, is within the measurement envelope **113** of the coordinate measurement machine **100**. For example, the robot **300** may deliver the workpiece **180** to a workholder **400** at the table **111** of the coordinate measuring machine **100**.

(149) In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies one or more parameters for operating the robot **300** to move and/or release the workpiece **180**. In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies a wait time parameter that quantitatively specifies a wait time between the time that the robot **300** grasps the workpiece **180**, and the time the robot **300** begins moving the workpiece, and/or a parameter that defines a safe position (specified as a set of coordinates in the coordinate system of the system **90**) for the effector **311** above or adjacent to the workpiece **180** to which the robot moves the effector **311** prior to grasping the workpiece, and/or an orientation of the effector at such a safe point prior to grasping the workpiece **180**.

(150) In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies a path through which the robot **300** moves the workpiece **180** in its grasp. For example, the ruleset **610** may specify that the robot **300** moves the workpiece **180** directly (e.g., in a straight line) from the point at which the robot **180** obtained the workpiece **180** to the point (the drop-off point) where the robot **300** is to deliver the workpiece **180**. In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies that the robot **300** is to move the workpiece **180** directly downward (e.g., in the -Z axis of the coordinate system of the inspection system **90**) after the workpiece **180** arrives at the drop-off point. That specification may quantitatively specify a fixed distance for that downward motion, or may specify that the downward motion continues until a threshold force of the workpiece **180** against a surface (e.g., the table of a coordinate measuring machine **100**, or a surface of a workholder **400**) is detected. In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies that the robot **300** is to move the workpiece **180** in a plane that is normal to the Z-axis (i.e., and X-Y plane) for a specified quantitative distance, or until a threshold

force of the workpiece **180** against a surface (e.g., a surface of a workholder **400**) is detected. (151) The method also includes step **550**, at which the method **500** delivers the workpiece **180** to the workholder **400**. In illustrative embodiments, a workpiece interface **420** of the workholder **400** is open to receive the workpiece **180**.

(152) In some embodiments, step **550** precedes step **540**. In other embodiments, such as when a workholder **400** is already positioned on a coordinate measuring machine, step **550** follows step **540** and the robot **300** delivers the workpiece **180** to the workholder **400**.

(153) Some workholders **400** may be able to receive a workpiece **180** from a robot **300** without operator intervention or assistance, but in such cases conventional workholders can only follow pre-programmed instructions, and lack the ability to adapt their actions to changing conditions. For example, conventional workholders cannot adapt their behavior to operate differently for different (e.g., non-identical) workpieces **180**. Sometimes, when consecutively receiving two workpieces **180** which workpieces **180** are not identical to one another, the workholder's operation for receiving (e.g., from the robot **300**) the first workpiece **180** may not be appropriate for receiving the second workpiece **180**, such as when the second workpiece is more delicate than the first workpiece and therefore requires a lower gripping pressure by the workholders than the first workpiece **180**, or such as when the second workpiece **180** has a different shape than the first workpiece **180**, and therefore requires that the workholder **400** grasp the second workpiece **180** in a location on the second workpiece **180** that is specific to that second workpiece **180**, and which would not be possible or viable for grasping the first workpiece **180**.

(154) In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies one or more parameters for operating the workholder **400** to receive, and/or hold, and/or release the workpiece **180**. In some embodiments, the ruleset **610** corresponding to the workpiece **180** specifies that the workpiece interface **420** is to be vibrated prior to clamping the workpiece **180** by the workpiece interface **180**. Such specification may include the duration of such vibration, and/or the intensity of such vibration.

(155) At step **560**, the inspection instrument (e.g., coordinate measuring machine **100**) inspects the workpiece **180** held in the workholder **400**.

(156) At step **570**, typically after the inspection instrument completes or terminates its inspection of the workpiece **180** held in the workholder **400**, the robot **300** retrieves the workpiece **180** from the workholder **400**. Some robots **300** may be able to retrieve a workpiece **180** from a workholder **400** without operator intervention, but in such cases conventional robots **300** can only follow pre-programmed instructions, and lack the ability to adapt their actions to changing conditions. For example, conventional robots cannot adapt their behavior to operate differently for different (e.g., non-identical) workpieces. Sometimes, when consecutively obtaining two workpieces **180** which workpieces **180** are not identical to one another, the robot's operation for obtaining the first workpiece **180** may not be appropriate for obtaining the second workpiece **180**.

(157) Moreover, the operation of the workholder **400** may depend on, or be correlated to, the specific workpiece **180**, such that the operation of the workholder **400** is different for each different workpiece. For example, each workpiece **180** may have specific corresponding requirements for how wide to open the jaws of the workholder **400**, how fast to open the workholder **400**, when to open the workholder relative to the motion or timing of the robot working to retrieve the workpiece **180** from the workholder **400**, to name but a few examples.

(158) At step **580**, after grasping the workpiece **180** when the workpiece **180** is within the grasp of and under control of the robot **300**, the method removes the workpiece from the workholder **400**, and from the measurement envelope **113** of the coordinate measuring machine **100**. In some embodiments, the robot **300** moves the workpiece **180** back to the workpiece storage apparatus **200**. In other embodiments, the robot **300** moves the workpiece **180** to a different storage location, or to a location specified for storing workpieces **180** that have failed inspection. In some embodiments, when a workpiece **180** fails inspection by the coordinate measuring machine **100**,

the robot **300** physically changes the workpiece **180**, for example by bending the workpiece **180**, crushing the workpiece **180**, or marking the workpiece **180**, to name but a few examples.

(159) At step **590**, the method determines whether there is at least one additional workpiece **180** to be inspected by the coordinate measuring machine. If not (“No”), then the method ends, but if so (“Yes”), then the method loops (step **591**) to step **530** to obtain the next workpiece **180** and its corresponding ruleset **610**. In some embodiments, the next workpiece **180** is non-identical to the previously-inspected workpiece **180**, and so parameters of the operation of the robot **300** and/or the workholder **400**, and/or the coordinate measuring machine **100**, may be automatically adjusted or adapted to customize the robot **300** and/or the workholder **400**, and/or the coordinate measuring machine **100** to perform the steps of the method for that next workpiece **180**.

(160) FIG. 5B is a flowchart of an embodiment of a method **551** of delivering a workpiece to a workholder **400**. Such a method may be employed, in some embodiments, at step **550**.

(161) Step **552** includes moving the workpiece **180** to a location near the workholder **400**. In illustrative embodiments, a workpiece interface **420** of the workholder **400** is open to receive the workpiece **180**. Illustrative embodiments moving the workpiece to a location near the work holder automatically using robot **300**. In some embodiments moving the workpiece to a location near the workholder includes moving the workpiece to a location above a workpiece interface **420** (which in some embodiments may be a clamp apparatus) such that the workpiece interface **420** is between the workpiece **180** and the ground, so that gravity acts on the workpiece **180** to as to pull the workpiece **180** towards the workpiece interface **420**.

(162) Step **553** includes aligning the workpiece **180** to the workpiece interface **420**, and step **554** includes disposing the workpiece **180** onto or into the workpiece interface **420**.

(163) Step **555** includes jiggling the workpiece interface **420** to cause the workpiece **180** to settle onto or into the workpiece interface **420**. In some embodiments, jiggling the workpiece interface **420** includes jiggling the entire workholder **420** while the workpiece **180** is in physical contact with the workholder **420**. In illustrative embodiments, step **555** is performed prior to closing the workpiece interface **420**, and separate from closing the workpiece interface **420**.

(164) The act of jiggling the workpiece interface **420** to cause the workpiece **180** to settle onto or into the workpiece interface **420** includes controllably and deliberately causing the workpiece interface **420** to physically oscillate or vibrate.

(165) In some embodiments, jiggling the workpiece interface **420** (e.g., causing the workpiece interface **420** to physically oscillate or vibrate, e.g., to jiggle the workpiece interface **420** and a workpiece **180** in the workpiece interface) includes activating an actuator coupled to the workpiece interface **420** and/or to the workholder **400**.

(166) For example, in embodiments of a workholder **400** that include a workholder actuator **413**, causing the workpiece interface **420** to physically oscillate or vibrate may include causing the workholder actuator **413** to cause the workholder interface **420** to perform a closing motion, followed by an opening motion before the workholder interface **420** secures the workpiece **180** to the workholder **400**. To that end, if the workholder actuator **413** includes an electric motor, causing the workpiece interface **420** to physically oscillate or vibrate may include controllably operating the electric motor to turn in a first direction to cause the workholder interface **420** to perform a closing motion, followed by controllably operating the electric motor to turn in a second direction to cause the workholder interface **420** to perform an opening motion. In an embodiment in which the workholder actuator **413** includes a fluid-drive actuator (e.g., a pneumatic actuator), causing the workpiece interface **420** to physically oscillate or vibrate may include controllably operating the fluid-drive actuator to apply a closing force to one or both of the first clamp arm **421** and the second clamp arm **422** to cause a closing motion, followed by controllably operating the fluid-drive actuator to apply an opening force to one or both of the first clamp arm **421** and the second clamp arm **422** to cause an opening motion.

(167) In some illustrative embodiments, a workholder actuator **413** for controllably closing the

workpiece interface **420** is configured to close the workpiece interface **420** by causing the workpiece interface to execute a closing motion, and is also configured to alternately open the workpiece interface **420** by causing the workpiece interface **420** to execute an opening motion, such that the actuator **413** is configured to alternately and controllably open and close the workpiece interface **420**. In such embodiments, the workholder **400** further includes a controller **411** to controllably vibrate (or jiggle) the workpiece interface **420** (to settle the workpiece into the workpiece interface **420**) by causing the actuator **413** to execute, alternately, a plurality of opening motions and closing motions, without changing or replacing the workpiece **180** between said opening motions and closing motions, and prior to closing the workpiece interface **420** to secure the workpiece.

(168) In embodiments of a workholder **400** that include a vibration actuator **430**, causing the workpiece interface **420** to physically oscillate or vibrate may include operating the vibration actuator **430** to transmit vibrations to the workpiece interface **420**. For example, in embodiment in which the vibration actuator **430** includes an electric motor, operating the vibration actuator **430** to transmit vibrations to the workpiece interface **420** may include causing the electric motor to turn. In an embodiment in which the vibration actuator **430** includes a fluid-driven actuator (e.g., a pneumatic actuator), operating the vibration actuator **430** to transmit vibrations to the workpiece interface **420** may include operating the fluid-driven actuator alternately to provide force in a first direction, followed by causing the fluid drive actuator to provide force in a second direction opposite the first direction.

(169) In illustrative embodiments, the act of jiggling the workpiece interface **420** is separate from the act of closing the workpiece interface **420**, and any vibration, oscillation or jiggling caused by or incidental to the act of closing the workpiece interface **420** to clamp or secure the workpiece **180** is separate from the act of jiggling the workpiece interface **420** at step **555**.

(170) In illustrative embodiments, step **555** is performed prior to closing the workpiece interface **420**, and separate from closing the workpiece interface **420**.

(171) Subsequent to jiggling the workpiece interface **420**, step **556** includes securing the workpiece **180** onto or into the workholder **400**. In illustrative embodiments, clamping the workpiece **180** onto or into the workholder **400** includes closing the workpiece interface **420** to clamp the workpiece **180** securely.

(172) FIG. **6A** schematically illustrates a ruleset **610** that includes and provides parameters (or “rules”) that specify one or more parameters or instructions for the operation of one or more inspection instruments of a workpiece inspection system **90**. Rulesets may also be referred-to as “parameter sets.” A ruleset **610** may include, for example, parameters for operating a robot **300** as part of an inspection system **90**, and/or parameters for operating a workholder **400** as part of an inspection system **90**, to name but a few examples. For example, a ruleset that includes parameters for operating a robot **300** may be referred to as a “robot ruleset.” A ruleset that includes parameters for operating a workholder **400** may be referred to as a “workholder ruleset.” FIG. **6A** schematically illustrates a ruleset **610** that may have a robot ruleset **611** and/or a workholder ruleset **612**. In general, a ruleset **610** may be provided in a JSON database file, or an XML file.

(173) A ruleset **610** may include one or more of the following parameters: A parameter specifying a vacuum pressure for a workholder that secures a workpiece by a vacuum force (e.g., a vacuum suction device); and/or A specified delay time between an event and an operation, such as a workholder operation, where the event may be a trigger signal or a previous workholder operation; and/or A parameter specifying that the workpiece interface **420** is to be vibrated prior to clamping the workpiece **180** by the workpiece interface **180**. Such a parameter may include the duration of such vibration, and/or the intensity of such vibration; and/or A parameter specifying a pre-grasp width of gripper opening gap **317** for obtaining (e.g., grasping or picking-up) the corresponding workpiece **180**, pursuant to which the system controller **91** causes the gripper to open to the specified pre-grasp width; and/or A parameter specifying a width of gripper opening gap **317** for

releasing (e.g., dropping or letting go of) the corresponding workpiece **180**, pursuant to which the system controller **91** causes the gripper to open to the specified release width; A parameter instructing the robot **300** to open the gripper to its maximum gap **317**, pursuant to which the system controller **91** controls the robot **300** to open the gripper **311** to its maximum width; and/or A parameter quantitatively specifying a gap which gap is less than the maximum gap of the gripper **311**, pursuant to which the system controller **91** controls the robot **300** to open the gripper **311** to the specified gap; and/or A parameter instructing the robot **300** to close the gripper **311** to its minimum gap; and/or A parameter specifying a wait time between the robot's effector arriving at a location of a workpiece **180** and a step of grasping said workpiece **180**, pursuant to which the system controller **91** causes the robot to delay grasping the workpiece until said wait time has elapsed; and/or Specification of a safe position above a workpiece **180** prior to grasping the workpiece **180** for delivery to a workpiece inspection machine **100**, the safe position specified in coordinates of the inspection system **90**, pursuant to which the system controller **91** causes the robot **300** to move the effector to the safe position prior to grasping the workpiece; Specification of the orientation of the robot's effector relative to the workpiece **180** prior to grasping the workpiece **180** for delivery to a workpiece inspection machine **100**, pursuant to which the system controller **91** causes the robot **300** or orient the effector to the specified orientation relative to the workpiece **180** prior to grasping the workpiece **180**; and/or Specification of a safe position above a workholder **400** prior to delivering the workpiece **180** to the workholder **400**, pursuant to which the system controller **91** causes the robot **300** to move the workpiece **180** to the safe position above the workholder **400** prior to delivering the workpiece **180** to the workholder **400**; and/or A parameter specifying an orientation of the effector holding a workpiece **180** prior to delivering the workpiece **180** to the workholder **400**, the orientation specified relative to the workholder **400** into which the workpiece **180** is to be placed, pursuant to which the system controller **91** causes the robot **300** to orient the workpiece to the specified orientation; and/or A parameter specifying a path pursuant to which the robot **300** moves the workpiece **180** directly to the workholder **400** in a direction normal to the workpiece interface until the workholder **400** applies to the workpiece **180** a specified quantitative force; and/or A parameter specifying a path pursuant to which the system controller **91** causes the robot **300** to move the workpiece **180** the workholder **400** in a direction in a plane, which plane is normal to an axis that is normal to the workpiece interface, until the workholder **400** applies to the workpiece **180** a specified quantitative force; and/or A parameter specifying that the workholder **400** should open the workpiece interface to its maximum workholder gap, pursuant to which the controller controls the workholder to open the workpiece interface to its maximum workholder gap; and/or A parameter specifying that the workholder **400** should close the workpiece interface to its minimum workholder gap, pursuant to which the controller controls the workholder to close the workpiece interface to its minimum workholder gap; and/or A parameter quantitatively specifying that the workholder **400** should open the workpiece interface to a specified distance, pursuant to which the controller controls the workholder to open the workpiece interface to the specified distance; and/or A parameter quantitatively specifying a closing force applied to the workpiece **180** by the workpiece interface, pursuant to which the controller controls the workholder to close its workpiece interface until said closing force is applied; and/or A parameter quantitatively specifying an opening force applied to the workpiece **180** by the workpiece interface, pursuant to which the controller controls the workholder to open its workpiece interface until said opening force is applied; and/or A parameter specifying a maximum closing speed for closing the workpiece interface, pursuant to which the controller controls the workholder to open the workpiece interface at a speed not greater than the specified maximum closing speed; and/or A parameter quantitatively specifying a closing delay time between (a) positioning of the workpiece **180** by a robot **300** in a specified position relative to the workpiece interface, and (b) closing of the workpiece interface to grasp the workpiece **180**, pursuant to which the controller controls the workholder to delay closing the workpiece interface until such closing

delay time has elapsed; and/or A parameter quantitatively specifying an opening delay time between (a) completion of an inspection operation by a workpiece inspection machine **100**, and (b) opening the workpiece interface to release the workpiece **180**, pursuant to which the controller controls the workholder to delay opening the workpiece interface until such opening delay time has elapsed,

(174) to name but a few examples.

(175) One or more instruments, or the operation of one or more instruments, of a system **90** under control of controller **91** may be customized pursuant to any one or more of the parameters described above.

(176) FIG. **6B** schematically illustrates several rulesets, each of which may be described generally as a ruleset **610**. FIG. **6B** includes several non-identical workpieces **681**, **682**, and **683**. Each non-identical workpiece **681**, **682**, **683** and **684** has a corresponding non-identical ruleset, in this embodiment rulesets **621**, **622**, and **623**, respectively. More specifically in this embodiment, workpiece **681** has a corresponding ruleset **621**; workpiece **682** has a corresponding ruleset **622**; and workpiece **683** has a corresponding ruleset **623**. Each ruleset **621-623** specifies operational parameters and/or instructions for controlling instruments of an inspection system **90** operating on the workpiece **681-683** corresponding to the ruleset.

(177) FIG. **6B** also schematically illustrates two families of parts, family **684** and family **685**, each of which includes a set of parts. In each family, each workpiece is associated with the same (or an identical) workpiece delivery ruleset (**624** and **625**, respectively, in this example) for customizing the configuration and/or the operation of at least one instruments of the set of instruments of a workpiece inspection system to move a workpiece and deliver the workpiece to a workholder. The workpieces in said set of workpieces may identical to one another, or may be non-identical to one another, as long as the customization or configuration of said set of instruments of a workpiece inspection system is performed pursuant to the same (or an identical) workpiece delivery ruleset.

(178) In an illustrative embodiment, family **684** includes a set having a plurality of parts. In the example of FIG. **6B**, the parts are numbered **686** and **687**. In some embodiments, the plurality of parts **686** and **687** are identical to one another, and in other embodiments, the plurality of parts **686** and **687** are non-identical to one another. In either case, each of the plurality of parts **686** and **687** is movable, by the robot **300**, pursuant to ruleset **624**.

(179) In an illustrative embodiment, family **685** includes a set having a plurality of parts. In the example of FIG. **6B**, the parts are numbered **688** and **689**. In some embodiments, the plurality of parts **688** and **689** are identical to one another, and in other embodiments, the plurality of parts **688** and **689** are non-identical to one another. In either case, each of the plurality of parts **688** and **689** is movable, by the robot **300**, pursuant to ruleset **625**.

(180) In operation, as part of obtaining a workpiece **180** at step **530**, an inspection system also obtains the ruleset for that workpiece **180**. For example, if an inspection system is operating on workpiece **681**, the system will obtain ruleset **621**; and if the inspection system is operating on workpiece **682**, the system will obtain ruleset **622**. As another example, if the inspection system is operating on either workpiece **686** or workpiece **687**, the system will obtain ruleset **624**. As another example, if the inspection system is operating on either workpiece **688** or workpiece **688**, the system will obtain ruleset **627**. To that end, the ruleset **610** may be stored in a memory **156** of a CMM controller; or in a memory of a computer **170** or computer **179**, to name but a few examples, each such ruleset **610** stored with information correlating the ruleset to a corresponding workpiece **180**.

(181) In some embodiments, the system (e.g., system controller **91**) recognizes or identifies each workpiece **180** obtained at step **530**, and in response identifies and retrieves the ruleset **610** corresponding to that workpiece **180**. For example, a system controller **91** may recognize or identify a workpiece **180** by imaging the workpiece **180** with a camera (e.g., CMM camera **141**) and assessing the image. For example, a system controller **91** may identify a workpiece **180** in an

image by assessing the image with one or more neural networks trained to recognize or identify a workpiece in an image. In other embodiments, workpiece inspection codes executing on a system controller **91** may specify each workpiece in a sequence of workpiece to be inspected, and contemporaneously identify and retrieve the ruleset corresponding to each such workpiece.

(182) According to the foregoing, some embodiments include a workpiece inspection system for sequentially delivering each workpiece of a plurality of workpieces, each workpiece being from a different family of workpieces, to a workholder. In some embodiments, the system includes a set of instruments, the set of instruments comprising a workpiece inspection instrument and a robot disposed to deliver to the workholder each workpiece of the plurality of workpieces, each workpiece of the plurality of workpieces being from a different family of a plurality of families of workpieces; a control system in control communication with the set of instruments of the workpiece inspection system, the control system configured to, for each workpiece: retrieve, from a plurality of workpiece delivery rulesets, a ruleset corresponding to the family of said workpiece, said corresponding ruleset comprising a set of parameters to automatically customize transfer of the workpiece to the workholder; and sequentially customize at least one instrument of the system and/or operation of at least one instrument of the system, according to any one or more of the parameters described above.

(183) For example, in some embodiments the system (via controller **91**) customizes at least one of (i) the configuration of the robot, and (ii) the operation of the robot, pursuant to the parameters of the corresponding ruleset; and subsequently (b) operates the robot to deliver said non-identical workpiece to the workholder.

(184) In some embodiments, the corresponding ruleset includes a parameter quantitatively specifying a closing delay between (a) positioning of the workpiece by the robot in a specified position relative to the workpiece interface, and (b) closing of the workpiece interface to grasp the workpiece, pursuant to which the control system the control system customizes the operation of the workholder to close the workpiece interface after passing of said closing delay.

(185) FIG. 7A and FIG. 7B each schematically illustrates an embodiment of a workholder **400**. Each embodiment has a workpiece interface **420** capable of an opening motion and a closing motion. Each embodiment is capable of operating in a plurality of distinct selectable, or specifiable, modes. Each distinct mode configures the workholder **400** to execute an associated, and distinct, set of workholder operations, which workholder operations are associated with an associated workpiece. Each of the embodiments of FIG. 7A and FIG. 7B has a multimode switch (e.g., **415**; **710**, described herein) configured to receive input to set the mode of the workholder **400**.

(186) In FIG. 7A, the multimode switch is implemented with a set of input signals provided to communications interface **415**. As schematically illustrated in FIG. 7C, the communications interface **415** is in electronic communication with the workholder processor **411**, and the electronic signals convey the selected mode to the workholder processor **411**. The workholder processor **411** is in control communication with workholder hardware **720**, and the workholder processor **411** stores, or has access to a memory that holds, instructions for controlling the workholder hardware **720** (e.g., workholder actuator **413** and/or vibration actuator **430**) to execute the workholder operations of the specified mode.

(187) In FIG. 7B, the multimode switch is a manually-operable switch **710**. In illustrative embodiments, the manually-operable switch **710** may be a may be a mechanical switch or an electronically operable switch. In operation, an operator (or, in some embodiments, robot **300**) may set the mode of the workholder **400** by physically setting the manually-operable switch **710** to one of a plurality of available mode positions. The manually-operable switch **710** may be referred-to as a “multi-mode” switch. Each mode position of the switch sets the workholder **400** to a distinct one of the modes.

(188) In some embodiments, as schematically illustrated in FIG. 7C, the manually-operable switch **710** is in electronic communication with the workholder processor **411**.

(189) In illustrative embodiments, the manually-operable switch sends a set of electronic signals defining the selected mode to the workholder processor **411**. The workholder processor **411** is in control communication with workholder hardware **720**, and the workholder processor **411** stores, or has access to a memory that holds, instructions for controlling the workholder hardware **720** to execute the workholder operations of the specified mode. In operation, the workholder processor **411** controls the workholder hardware **720** to execute the workholder operations of the specified mode.

(190) In operation, each workholder **400** is configured to execute workholding operations specified by the workholding mode to which it has been set. In some embodiments, a workholding mode is an embodiment of a ruleset **610**, and in some embodiments a workholding mode is a subset of a ruleset **610**, wherein the parameters of the workholding mode are configured to be read by workholder processor **411** (which workholder processor is part of and integral to the workholder **400**), for example from a memory within or accessible by the workholder **400** and/or workholder processor **411**, which workholder processor **411** then controls the workholder **400** and its operations accordingly by its control of workholder hardware **720**.

(191) FIG. **9B** is a flowchart of an embodiment of execution of workholder operations of specified mode.

(192) In illustrative embodiments, the workholder **400** is configured to execute such workholding operations autonomously. For example, in illustrative embodiments, once a workholder **400** is receives specification of, or is set to, a specific workholding mode, the workholder **400** is configured to autonomously execute the workholding operations specified by the workholding mode.

(193) In some embodiments, the workholder **400** is configured to pause between execution of at least one workholding operation (e.g., workholding operation N) and a subsequent workholding operation (e.g., workholding operation N+1), and to execute the subsequent workholding operation (e.g., workholding operation N+1) after passage of a pre-determined amount of time, and/or after receipt of a trigger signal (e.g., trigger signal N+1). In some embodiments, such a trigger signal may be provided to the workholder **400** from a client computer **750** via electronic signal via communications interface **415**. In accordance with the method of FIG. **9B**, a pause or trigger (e.g., trigger **952**; trigger **954**; trigger **956**; trigger **958**) may be included between execution of the first workholder operation and the execution of the second workholder operation; and/or between execution of the second workholder operation and the execution of the third workholder operation; and/or between execution of the third workholder operations and the execution of the fourth workholder operation, to provide but a few examples.

(194) FIG. **8A** and FIG. **8B** schematically illustrate and embodiment of a workholder **400** (which may be, for example, a workholder **400** with a multimode switch) in operation. This embodiment may be described as being in “chicken egg holding” mode, in which the workholder is configured to and operating to hold a chicken egg as workpiece **180**. In this illustrative embodiment, the mode (the “chicken egg holding” mode) specified the following parameters: Unclamping width=50 mm; Clamping force=0.1 Newton; Delay time from placement of workpiece **180** into workholder=0.0 seconds; and Clamping width=25 mm.

(195) Another type of egg may have a distinct workholding mode. For example, an ostrich egg would have different workholding specification than a chicken egg, due at least to the difference in size, even though a chicken egg and an ostrich egg are both eggs. In other words, not all eggs fall within the same family of workpieces and not all eggs have the same workholding mode.

Continuing with this example (“ostrich egg holding” mode), the workholder **400** is configured to and operating to hold an ostrich egg as workpiece **180**. In this illustrative embodiment, the mode (the “ostrich egg holding” mode) specified the following parameters: Unclamping width=125 mm; Clamping force=0.1 Newton; Delay time from placement of workpiece **180** into workholder=1.0 seconds; and Clamping width=99 mm.

(196) Operation of the workholder **400** in such a mode may include the following workholder operations in the following order: (i) opening the workpiece interface **425** to the unclamping width and, after receiving the workpiece in the workpiece interface **425**, and then (ii) closing the workpiece interface **425** to the clamping width. The action of closing the workpiece interface **425** to the clamping width in this embodiment is executed without a delay (i.e., the delay is 0.0 seconds), but in other embodiments the action of closing the workpiece interface **425** to the clamping width occurs after a delay of greater than 0.0 seconds (i.e., a non-zero delay), as specified in the selected mode. In some embodiments, the action of closing the workpiece interface **425** to the clamping width occurs only after receipt by the workholder **400** of a trigger signal, which may be referred-to as a closing action trigger.

(197) FIG. **8C** and FIG. **8D** schematically illustrate another embodiment of a workholder **400** (which may be, for example, a workholder **400** with a multimode switch) in operation. This embodiment may be described as being in “brick holding” mode, in which the workholder is configured to hold, and is operating to hold, a brick as workpiece **180**. A brick is a different type of workpiece than the egg of FIG. **8A**, and has a different workholding specification.

(198) FIG. **9A** is a flowchart of an embodiment of a method **900** of operation of an autonomous workholder during sequential inspection of a plurality of workpieces, wherein each workpiece has a holding specification distinct from the respective holding specifications of other workpieces of the plurality of workpieces.

(199) Step **910** includes providing a plurality of distinct, selectable pre-defined modes. Each such mode specifies a plurality of holding parameters corresponding to the holding specification of a corresponding workpiece of the plurality of workpieces.

(200) Step **920** includes providing the workholder **400** to hold each workpiece during inspection by an inspection instrument of an inspection system. The same workholder **400** holds each workpiece during the sequential inspection of a plurality of workpieces. The workholder **400** is configured to autonomously execute operations specified by the parameters of each of the pre-defined modes. Illustrative embodiments of such a workholder **400** are described above and schematically illustrated in figures.

(201) Step **930** includes receiving, at the workholder **400**, specification of a pre-defined mode from the plurality of pre-defined modes, which pre-defined mode may be referred-to as the specified pre-defined mode.

(202) Step **940** includes causing the workholder **400** to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode.

(203) FIG. **9B** is a flowchart of an embodiment of a method **940** of a workholder **400** sequentially executing a plurality of workholding operations.

(204) Step **942** includes executing a first workholder operation pursuant to a first parameter of the specified pre-defined mode. The workholder operation may be any one of the workholding operations described herein, including for example opening the workholder's workpiece interface pursuant to a parameter of the corresponding workholding mode; closing the workholder's workpiece interface pursuant to a parameter of the corresponding workholding mode (e.g., a specified clamping force applied to the workpiece); and/or jiggling (or “vibrating”) the workholder's workpiece interface pursuant to a time duration parameter and/or a vibration intensity parameter of the corresponding workholding mode; securing a workpiece with a vacuum pressure pursuant to a parameter of the corresponding workholding mode.

(205) Step **944** includes executing a second workholder operation pursuant to a second parameter of the specified pre-defined mode. In some embodiments, step **944** (or preceding step **942**) may include a delay of pre-determined duration between execution of step **942** and step **944**. In some embodiments, the method pauses after execution of step **942**, and executes step **944** only after, and in response to, receipt by the workholder **400** of a trigger signal triggering step **944**. The second workholder operation may be one of the workholder operations described above.

(206) Step **946** includes executing a third workholder operation pursuant to a third parameter of the specified pre-defined mode. In some embodiments, step **946** (or preceding step **944**) may include a delay of pre-determined duration between execution of step **944** and step **946**. In some embodiments, the method pauses after execution of step **944**, and executes step **946** only after, and in response to, receipt by the workholder **400** of a trigger signal triggering step **946**. The third workholder operation may be one of the workholder operations described above.

(207) Step **948** includes executing a fourth workholder operation pursuant to a fourth parameter of the specified pre-defined mode. In some embodiments, step **948** (or preceding step **946**) may include a delay of pre-determined duration between execution of step **948** and step **946**. In some embodiments, the method pauses after execution of step **946**, and executes step **948** only after, and in response to, receipt by the workholder **400** of a trigger signal triggering step **946**. The fourth workholder operation may be one of the workholder operations described above.

REFERENCE NUMBERS

(208) Reference numbers used herein include the following: **90**: Workpiece inspection system; **91**: Workpiece inspection system controller (or “computer implemented controller”); **92**: Ruleset database; **100**: Coordinate measuring machine; **101**: Floor; **102**: Environment; **110**: Base; **111**: Table; **112**: Plane; **113**: Measurement space (or measurement envelope); **115**: Probe rack; **120**: Moveable features; **121**: Bridge legs; **122**: Table scale; **123**: Bridge; **124**: Bridge scale; **125**: Carriage; **126**: Spindle; **127**: Spindle scale; **128**: Bearing; **130**: Arm; **131**: Moveable joint; **132**: Rotary encoder; **140**: Measuring sensor; **141**: Camera; **142**: Environmental sensor; **150**: Control system; **151**: Bus; **152**: Communications interface; **153**: Motion Controller; **154**: Measurement analyzer; **155**: Sensor input; **156**: Memory; **157**: Computer processor; **160**: User interface; **161**: X-axis controls; **162**: Y-axis controls; **163**: Z-axis controls; **165**: Camera motion controls; **166**: Camera focus control; **167**: Camera record control; **170**: Host computer; **171**: Screen; **172**: Keyboard; **173**: Mouse; **174**: Computer memory; **175**: Memory interface/communications port; **176**: Communication link; **178**: Network; **179**: Computer; **180**: Workpiece; **181**: Geometry; **182**: Edge; **183**: Corner; **184**: Flat surface; **185**: Curved surface; **186**: Cavity; **187**: Inside angle; **188**: Waviness; **189**: Surface finish; **190**: Jogbox; **191**: Cable; **200**: Workpiece storage apparatus; **201**: Storage container; **202**: Storage plate surface; **203**: Storage plate; **300**: Robot; **301**: Robot base; **302**: Robot arm; **303**: Distal end of robot arm; **311**: Robot gripper; **314**: First gripper finger; **315**: Second gripper finger; **317**: Robot gripper gap; **340**: Robot end effector (e.g., gripper, etc.); **379**: Robot control computer; **390**: Robot control interface; **400**: Workholder; **410**: Workholder base; **411**: Workholder processor; **413**: Workholder actuator; **415**: Communications interface; **416**: Fluid Interface; **420**: Workpiece interface; **421**: First clamp arm; **422**: Second clamp arm; **425**: Controllable workholder gap; **430**: Vibration actuator; **450**: Post; **451**: Slide surface; **460**: Corner guide; **461** and **462**: surfaces; **470**: End stop; **471**: Porch; **472**: Wall of end stop; **480**: Workpiece aperture; **481**: Edge of workpiece aperture; **710**: Multimode switch; **720**: Workholder hardware (e.g., workholder actuator; vibration actuator); **730**: Power source for workholder; **750**: PC Client (e.g., control system **150**; host computer **170**); **952**: First workholder operation trigger; **954**: First workholder operation trigger; **956**: First workholder operation trigger; **958**: First workholder operation trigger.

(209) Various embodiments may be characterized by the potential claims listed in the paragraphs following this paragraph (and before the actual claims provided at the end of this application). These potential claims form a part of the written description of this application. Accordingly, subject matter of the following potential claims may be presented as actual claims in later proceedings involving this application or any application claiming priority based on this application. Inclusion of such potential claims should not be construed to mean that the actual claims do not cover the subject matter of the potential claims. Thus, a decision to not present these potential claims in later proceedings should not be construed as a donation of the subject matter to the public.

(210) Without limitation, potential subject matter that may be claimed (prefaced with the letter “P” so as to avoid confusion with the actual claims presented below) includes:

(211) P1. A method of operating a workholder having a workpiece interface configured to controllably clamp a workpiece, the method comprising:

(212) placing the workpiece into the workpiece interface, the workpiece interface being open to receive the workpiece; prior to closing the workpiece interface, and separate from closing the workpiece interface, vibrating the workpiece interface to settle the workpiece into the workpiece interface; and subsequent to vibrating the workpiece interface closing the workpiece interface to secure the workpiece to the workholder.

P2. The method of P1, further comprising delivering, by a robot, the workpiece proximate to the workpiece interface, and pausing for a pre-established amount of time prior to placing the workpiece into the workpiece interface.

P3. The method of any of P1-P2, wherein: the workholder comprises a workholder actuator operably coupled to the workpiece interface to controllably close the workpiece interface, and vibrating the workpiece interface comprises operating the workholder actuator without closing the workpiece interface.

P4. The method of any of P1-P3, wherein: the workholder comprises a workholder actuator operably coupled to the workpiece interface to controllably open the workpiece interface with an opening motion, and to controllably close the workpiece interface with a closing motion, and vibrating the workpiece interface to settle the workpiece into the workpiece interface comprises alternating the workholder actuator between an opening motion and a closing motion.

P5. The method of any of P1-P4, wherein the workholder comprises a vibration actuator in addition to the workholder actuator, and vibrating the workpiece interface comprises operating the vibration actuator without closing the workpiece interface; and closing the workpiece interface comprises operating workholder actuator separately from operating the vibration actuator.

P6. The method of any of P1-P5, wherein the workpiece interface comprises a substrate defining a substrate plane and a set of guides, each guide of the set of guides having a surface extending away from the substrate plane, and disposed and configured to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

P7. The method of P6, wherein the set of guides includes a tapered post, the tapered post having a slide surface angled to intersect the substrate plane at an intersection such that the slide surface and the substrate plane form an obtuse angle at said intersection, the slide surface disposed and configured to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

P8. The method of P7, wherein the workpiece defines a workpiece aperture having an edge, and the tapered post is configured to pass through the workpiece aperture and the slide surface is disposed and configured to engage the edge to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

P9. The method of P6, wherein the set of guides includes a corner guide having two surfaces, the two surfaces extending away from the substrate plane, the two surfaces disposed at an angle to one another, so as to guide the workpiece onto the substrate in two corresponding directions in response to vibrating the workpiece interface.

P10. The method of P6, wherein the set of guides includes a pair of corner guides, each corner guide of the pair of corner guides having two tapered surfaces, the two tapered surfaces disposed at an angle to one another, so as to guide the workpiece onto the substrate in two corresponding directions in response to vibrating the workpiece interface, wherein the pair of corner guides cooperate, in response to vibrating the workpiece interface, to automatically center the workpiece in at least one axis on the substrate plane.

P11. The method of any of P1-P10, wherein the workpiece interface comprises a substrate defining a substrate plane, and having a set of end stops, each end stop of the set of end stops having a set of

end stop faces extending in a direction away from the substrate plane, each end stop of the set of end stops disposed and configured to constrain the workpiece while the workpiece is clamped by the workpiece interface.

P12. A workholder to receive and securely hold a workpiece, the workholder comprising: a base; a workpiece interface coupled to the base, the workpiece interface configured to open into an open configuration to receive a workpiece, and configured to close into a closed position to secure the workpiece; and a set of actuators, at least one actuator of the set of actuators operably coupled to the workpiece interface to controllably close the workpiece interface; and at least one actuator of the set of actuators operably coupled to the workholder to controllably vibrate the workpiece interface independent of closing the workpiece interface.

P13. The workholder of P12, wherein the set of actuators includes a workholder actuator, said workholder actuator is operably coupled to the workpiece interface and configured: to controllably open the workpiece interface with an opening motion, and to controllably close the workpiece interface with a closing motion; and, separately from closing the workpiece interface, to controllably vibrate the workpiece interface by alternately causing an opening motion and a closing motion without clamping the workpiece into the workpiece interface.

P14. The workholder of any of P12-P13, wherein the set of actuators includes: a workholder actuator operably coupled to the workpiece interface to controllably close the workpiece interface; and a vibration actuator operably coupled to the workholder to controllably vibrate the workpiece interface independent of closing the workpiece interface. the workholder actuator is a distinct actuator from the vibration actuator and the workholder actuator is controllably operable to close the workpiece interface independently of control of the vibration actuator to vibrate the workpiece interface.

P15. The workholder of any of P12-P14, wherein the workpiece interface comprises: a substrate defining a substrate plane; and a set of tapered guides, each tapered guide of the set of tapered guides having a surface extending away from the substrate plane, and disposed and configured to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

P16. The workholder of P15, wherein the set of tapered guides includes a tapered post, the tapered post having an slide surface, the slide surface disposed relative to the substrate plane such that the slides surface intersects the substrate plane such that the slide surface and substrate plane form an obtuse angle, the slide surface disposed and configured to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

P17. The workholder of P15, wherein the workpiece defines a workpiece aperture having an edge, and the tapered post is configured to pass through the workpiece aperture and the slide surface is disposed and configured to engage the edge to guide the workpiece onto the substrate in response to vibrating the workpiece interface.

P18. A workholder to receive and securely hold a workpiece, the workholder comprising: means for receiving a workpiece and controllably closing to clamp the workpiece, said means being workpiece interface means; means for controllably closing the workpiece interface means; and means for controllably vibrating the workpiece interface means to settle the workpiece into the workpiece interface means prior to closing to clamp the workpiece.

P19. The workholder of P18, wherein the means for controllably closing the workpiece interface means is distinct from, and is controllable independently of, the means for controllably vibrating the workpiece interface means.

P31. A method of operating a workholder for inspecting a workpiece, said workpiece having a distinct holding specification, the method comprising: providing a pre-defined mode specifying a plurality of holding parameters corresponding to a holding specification of the workpiece of the plurality of workpieces; providing the workholder to hold the workpiece during inspection by an inspection instrument of an inspection system, the workholder configured to autonomously execute operations specified by the parameters of the pre-defined mode; and

causing the workholder to autonomously execute holding operations pursuant to the parameters of the pre-defined mode.

P41. A method of sequentially delivering each workpiece of a plurality of workpieces, each workpiece being from a different family of workpieces, to a workholder to hold the workpiece during inspection by an inspection instrument of an inspection system, the inspection system comprising the inspection instrument, a robot having an end effector configured to grasp each workpiece, the robot disposed to move each workpiece directly to the measurement volume of the inspection instrument, and a control system in control communication with the robot, the method comprising: providing a plurality of non-identical workpieces for inspection by the workpiece inspection system, each workpiece of the plurality of workpieces being from a different family of a plurality of families of workpieces; providing a plurality of workpiece delivery rulesets, each ruleset of the plurality of workpiece delivery rulesets corresponding, respectively, to a family from the plurality of families of workpieces; and, for each such non-identical workpiece: retrieving, from the plurality of workpiece delivery rulesets, a ruleset corresponding to the family of said workpiece, said corresponding ruleset comprising a set of parameters to automatically (a) customize at least one of: (i) the configuration of the workholder, and (ii) the operation of the workholder, pursuant to the parameters of the corresponding ruleset; and subsequently (b) operate the workholder to perform at least one of clamping the workpiece; holding the workpiece during inspection of the workpiece by the inspection instrument; and releasing the workpiece subsequent to inspection of the workpiece by the inspection instrument.

P51: A method of operating a workholder during sequential inspection of a plurality of workpieces, each workpiece having a holding specification distinct from the respective holding specifications of other workpieces of the plurality of workpieces, the method comprising: providing a plurality of distinct, selectable pre-defined modes, each such mode specifying a plurality of holding parameters corresponding to the holding specification of a corresponding workpiece of the plurality of workpieces, each holding parameter of the plurality of holding parameters corresponding to a corresponding workholding operation of a plurality of workholding operations; providing the workholder to hold each workpiece during inspection by an inspection instrument of an inspection system, the workholder configured to autonomously execute workholding operations pursuant as specified by the holding parameters of each of the pre-defined modes; receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes, said pre-defined mode being a specified pre-defined mode; and causing the workholder to autonomously execute the plurality of workholding operations pursuant to the parameters of the specified pre-defined mode.

P52. The method of P51, wherein: the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and wherein receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving a set of mode control signals pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

P53. The method of P51, wherein: the workholder includes a communications interface in communication with a control computer and in control communication with workholder hardware; and wherein receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving a set of mode control signals from the control computer to the communications interface.

P54. The method of any of P51-P53, wherein: the specified pre-defined mode comprises a parameter defining a receiving width of a workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to open the workpiece interface to the receiving width.

P55. The method of any of P51-P54, wherein: the specified pre-defined mode comprises a

parameter defining a clamping width of a workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to close the workpiece interface to the clamping width.

P56. The method of any of P51-P55, wherein: the specified pre-defined mode comprises a parameter defining a clamping force to be applied by a workpiece interface of the workholder to the workpiece; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to close the workpiece interface to apply the clamping force.

P57. The method of any of P51-P56, wherein: the specified pre-defined mode comprises a parameter defining a vacuum pressure of a workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to apply the vacuum pressure to the workpiece.

P58. The method of any of P51-P57, wherein: the specified pre-defined mode comprises a parameter defining a voltage applied to a workholder actuator of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to apply said voltage to the workholder actuator of the workholder.

P59. The method of any of P51-P58, wherein: the specified pre-defined mode comprises a parameter defining a time duration to jiggle the workpiece interface upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to jiggle the workpiece for the specified time duration.

P60. The method of any of P51-P59, wherein: the specified pre-defined mode comprises a parameter defining a vibration intensity at which to jiggle the workpiece interface upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to jiggle the workpiece at the specified vibration intensity.

P71. A workholder apparatus configured for sequentially holding each workpiece of a plurality of workpieces, each workpiece having a distinct holding specification from the respective holding specifications of other workpieces of the plurality of workpieces, the workholder comprising: a workpiece interface controllable to open to receive the workpiece in an open configuration, and to close to grasp the workpiece in a closed configuration; an actuator integral to the workholder and mechanically coupled to the workpiece interface; and a control circuit integral to the workholder, the control circuit configured to (1) receive specification of a pre-defined mode from a plurality of selectable pre-defined modes, each such mode specifying a plurality of holding parameters corresponding to the holding specification of a workpiece of the plurality of workpieces, said pre-defined mode being a specified pre-defined mode, and to (2) autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode.

P72. The workholder apparatus of P71, wherein: the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and wherein to receive, at the control circuit, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving at the control circuit control signals pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

P73. The workholder apparatus of any of P71-P72, wherein: the workholder includes a communications interface and in control communication with workholder hardware; and wherein to receive, at the control circuit, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving a set of mode control signals from the control computer to the

communications interface.

P74. The workholder apparatus of any of P71-P73, wherein: the specified pre-defined mode comprises a parameter defining a clamping width of the workpiece interface of the workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to close the workpiece interface to the clamping width.

P75. The workholder apparatus of any of P71-P74, wherein: the specified pre-defined mode comprises a parameter defining a time duration to jiggle the workpiece interface and a vibration intensity at which to jiggle the workpiece interface, upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute holding operations pursuant to the parameters of the specified pre-defined mode comprises causing the workholder to jiggle the workpiece at the specified vibration intensity for the specified time duration.

P81. A non-transitory computer readable medium having non-transient computer-executable code, the non-transient computer-executable code for controlling a workholder for autonomously executing holding operations pursuant to parameters of a specified pre-defined mode, the computer-executable code comprising: code for causing the workholder to selectively execute workholding operations of each mode of a plurality of distinct, selectable pre-defined modes, each such mode specifying a plurality of holding parameters corresponding to the holding specification of a workpiece of a plurality of workpieces; code for receiving, at a control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes, said pre-defined mode being a specified pre-defined mode; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode.

P82. The non-transitory computer readable medium of P81, wherein the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and wherein code for receiving, at the control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises code for receiving specification of a pre-defined mode pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

P83. The non-transitory computer readable medium of any of P81-P82, wherein the workholder includes a communications interface, and code for receiving, at the control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises code for receiving a set of mode control signals from a control computer at the communications interface.

P84. The non-transitory computer readable medium of any of P81-P84, wherein the specified pre-defined mode comprises a parameter defining a receiving width of a workpiece interface of the workholder; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode comprises code for causing the workholder to open the workpiece interface to the receiving width.

P85. The non-transitory computer readable medium of any of P81-P85, wherein the specified pre-defined mode comprises: a first parameter defining a receiving width of a workpiece interface of the workholder; and a second parameter defining a clamping width of a workpiece interface of the workholder; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode comprises: code for causing the workholder to open the workpiece interface to the receiving width; and code for receiving a closing trigger signal subsequent to opening the workpiece interface to the receiving width; and code for causing the workholder to close the workpiece interface to the clamping width in response to receipt of the closing trigger signal.

(213) Various embodiments of this disclosure may be implemented at least in part in any

conventional computer programming language. For example, some embodiments may be implemented in a procedural programming language (e.g., “C”), or in an object-oriented programming language (e.g., “C++”), or in Python, R, Java, LISP, or Prolog. Other embodiments of this disclosure may be implemented as preprogrammed hardware elements (e.g., application specific integrated circuits, FPGAs, and digital signal processors), or other related components.

(214) In an alternative embodiment, the disclosed apparatus and methods may be implemented as a computer program product for use with a computer system. Such implementation may include a series of computer instructions fixed either on a tangible medium, such as a non-transient computer readable medium (e.g., a diskette, CD-ROM, ROM, FLASH memory, or fixed disk). The series of computer instructions can embody all or part of the functionality previously described herein with respect to the system.

(215) Those skilled in the art should appreciate that such computer instructions can be written in a number of programming languages for use with many computer architectures or operating systems. Furthermore, such instructions may be stored in any memory device, such as semiconductor, magnetic, optical or other memory devices, and may be transmitted using any communications technology, such as optical, infrared, microwave, or other transmission technologies.

(216) Among other ways, such a computer program product may be distributed as a removable medium with accompanying printed or electronic documentation (e.g., shrink wrapped software), preloaded with a computer system (e.g., on system ROM or fixed disk), or distributed from a server or electronic bulletin board over the network (e.g., the Internet or World Wide Web). Of course, some embodiments of this disclosure may be implemented as a combination of both software (e.g., a computer program product) and hardware. Still other embodiments of this disclosure are implemented as entirely hardware, or entirely software.

(217) Computer program logic implementing all or part of the functionality previously described herein may be executed at different times on a single processor (e.g., concurrently) or may be executed at the same or different times on multiple processors and may run under a single operating system process/thread or under different operating system processes/threads. Thus, the term “computer process” refers generally to the execution of a set of computer program instructions regardless of whether different computer processes are executed on the same or different processors and regardless of whether different computer processes run under the same operating system process/thread or different operating system processes/threads.

(218) The embodiments described above are intended to be merely exemplary; numerous variations and modifications will be apparent to those skilled in the art. All such variations and modifications are intended to be within the scope of the present disclosure as defined in any appended claims.

Claims

1. A method of operating a workholder during sequential inspection of a plurality of workpieces, the workholder having a workpiece interface, each workpiece having a holding specification distinct from the respective holding specifications of other workpieces of the plurality of workpieces, the method comprising: providing a plurality of distinct, selectable pre-defined modes, each such mode specifying a set of holding parameters corresponding to the holding specification of a corresponding workpiece of the plurality of workpieces, each holding parameter of the set of holding parameters corresponding to a corresponding workholding operation, the set of holding parameters comprising at least one of: a parameter quantitatively specifying a maximum closing speed for closing the workpiece interface; a parameter quantitatively specifying a closing delay time between (a) positioning of the workpiece by a robot in a specified position relative to the workpiece interface, and (b) closing of the workpiece interface to grasp the workpiece; a parameter quantitatively specifying an opening delay time between (a) completion of an inspection operation by a workpiece inspection machine and (b) opening the workpiece interface to release the

workpiece; a parameter defining a time duration to jiggle the workpiece interface subsequent to receipt of the workpiece at the workholder; and/or a parameter defining a vibration intensity at which to jiggle the workpiece interface subsequent to receipt of the workpiece at the workholder; providing the workholder to hold each workpiece during inspection by an inspection instrument of an inspection system, the workholder configured to autonomously execute workholding operations pursuant to at least one of the holding parameters of the set of holding parameters of each of the pre-defined modes; receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes, said pre-defined mode being a specified pre-defined mode; and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode.

2. The method of claim 1, wherein: the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and wherein receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving a set of mode control signals pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

3. The method of claim 1, wherein: the workholder includes a communications interface in communication with a control computer and in control communication with workholder hardware; and wherein receiving, at the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving a set of mode control signals from the control computer to the communications interface.

4. The method of claim 1, wherein: the set of holding parameters comprises the parameter specifying a maximum closing speed for closing the workpiece interface, and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to close the workpiece interface at a speed not greater than the specified maximum closing speed.

5. The method of claim 1, wherein: the specified pre-defined mode comprises a parameter defining a clamping width of a workpiece interface of the workholder; and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to close the workpiece interface to the clamping width.

6. The method of claim 1, wherein: the set of holding parameters comprises the parameter quantitatively specifying the closing delay time between (a) positioning of the workpiece by a robot in a specified position relative to the workpiece interface, and (b) closing of the workpiece interface to grasp the workpiece, and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to delay closing the workpiece interface until such closing delay time has elapsed.

7. The method of claim 1, wherein: the set of holding parameters comprises parameter quantitatively specifying the opening delay time between (a) completion of an inspection operation by a workpiece inspection machine and (b) opening the workpiece interface to release the workpiece, and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to delay opening the workpiece interface until such opening delay time has elapsed.

8. The method of claim 1, wherein: the specified pre-defined mode comprises a parameter defining a voltage applied to a workholder actuator of the workholder; and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to

apply said voltage to the workholder actuator of the workholder.

9. The method of claim 1, wherein: the specified pre-defined mode comprises the parameter defining a time duration to jiggle the workpiece interface upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to jiggle the workpiece for the specified time duration.

10. The method of claim 1, wherein: the specified pre-defined mode comprises the parameter defining a vibration intensity at which to jiggle the workpiece interface upon receipt of the workpiece at workholder; and causing the workholder to autonomously execute each workholding operation of the set of workholding operations pursuant to the holding parameters of the specified pre-defined mode comprises causing the workholder to jiggle the workpiece at the specified vibration intensity.

11. A workholder apparatus configured for sequentially holding each workpiece of a plurality of workpieces, each workpiece having a distinct holding specification from the respective holding specifications of other workpieces of the plurality of workpieces, the workholder comprising: a workpiece interface controllable to open to receive the workpiece in an open configuration, and to close to grasp the workpiece in a closed configuration; an actuator integral to the workholder and mechanically coupled to the workpiece interface; and a control circuit integral to the workholder, the control circuit configured to (1) receive specification of a pre-defined mode from a plurality of selectable pre-defined modes, each such mode specifying a set of holding parameters corresponding to the holding specification of a workpiece of the plurality of workpieces, said pre-defined mode being a specified pre-defined mode, and to (2) autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode, wherein the holding parameters comprise at least one of: a parameter specifying a maximum closing speed for closing the workpiece interface, pursuant to which a controller controls the workholder to open the workpiece interface at a speed not greater than the specified maximum closing speed; a parameter quantitatively specifying a closing delay time between (a) positioning of the workpiece by a robot in a specified position relative to the workpiece interface, and (b) closing of the workpiece interface to grasp the workpiece, pursuant to which the controller controls the workholder to delay closing the workpiece interface until such closing delay time has elapsed; a parameter quantitatively specifying an opening delay time between (a) completion of an inspection operation by a workpiece inspection machine and (b) opening the workpiece interface to release the workpiece, pursuant to which the controller controls the workholder to delay opening the workpiece interface until such opening delay time has elapsed; a parameter defining a time duration to jiggle the workpiece interface upon receipt of the workpiece at workholder, pursuant to which the workholder jiggles the workpiece for the specified time duration; and/or a parameter defining a vibration intensity at which to jiggle the workpiece interface upon receipt of the workpiece at workholder, pursuant to which the workholder jiggles the workpiece at the specified vibration intensity.

12. The workholder apparatus of claim 11, wherein: the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and wherein to receive, at the control circuit, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving at the control circuit control signals pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

13. The workholder apparatus of claim 11, wherein: the workholder includes a communications interface and in control communication with workholder hardware; and wherein to receive, at the control circuit, specification of a pre-defined mode from the plurality of pre-defined modes comprises receiving a set of mode control signals from the control computer to the communications interface.

14. The workholder apparatus of claim 11, wherein: the specified pre-defined mode comprises a

parameter defining a clamping width of the workpiece interface of the workholder.

15. The workholder apparatus of claim 11, wherein: the specified pre-defined mode comprises the parameter defining a time duration to jiggle the workpiece interface and a vibration intensity at which to jiggle the workpiece interface, upon receipt of the workpiece at workholder.

16. A non-transitory computer readable medium having non-transient computer-executable code, the non-transient computer-executable code for controlling a workholder for autonomously executing holding operations pursuant to parameters of a specified pre-defined mode, the computer-executable code comprising: code for causing the workholder to selectively execute workholding operations of each mode of a plurality of distinct, selectable pre-defined modes, each such mode specifying a set of holding parameters corresponding to the holding specification of a workpiece of a plurality of workpieces the holding parameters comprising at least one of: a parameter specifying a maximum closing speed for closing the workpiece interface, pursuant to which a controller controls the workholder to open the workpiece interface at a speed not greater than the specified maximum closing speed; a parameter quantitatively specifying a closing delay time between (a) positioning of the workpiece by a robot in a specified position relative to the workpiece interface, and (b) closing of the workpiece interface to grasp the workpiece, pursuant to which the controller controls the workholder to delay closing the workpiece interface until such closing delay time has elapsed; a parameter quantitatively specifying an opening delay time between (a) completion of an inspection operation by a workpiece inspection machine and (b) opening the workpiece interface to release the workpiece, pursuant to which the controller controls the workholder to delay opening the workpiece interface until such opening delay time has elapsed; a parameter defining a time duration to jiggle the workpiece interface upon receipt of the workpiece at workholder, pursuant to which the workholder jiggles the workpiece for the specified time duration; and/or a parameter defining a vibration intensity at which to jiggle the workpiece interface upon receipt of the workpiece at workholder, pursuant to which the workholder jiggles the workpiece at the specified vibration intensity; code for receiving, at a control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes, said pre-defined mode being a specified pre-defined mode; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode.

17. The non-transitory computer readable medium of claim 16, wherein the workholder includes a manually-operable mode switch having a plurality of distinct configurations, each configuration causing the manually-operable switch to establish the workholder into a corresponding distinct one of the pre-defined modes; and wherein code for receiving, at the control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises code for receiving specification of a pre-defined mode pursuant to manually setting the manually-operable mode switch to one of a plurality of configurations.

18. The non-transitory computer readable medium of claim 16, wherein the workholder includes a communications interface, and code for receiving, at the control circuit integral to the workholder, specification of a pre-defined mode from the plurality of pre-defined modes comprises code for receiving a set of mode control signals from a control computer at the communications interface.

19. The non-transitory computer readable medium of claim 16, wherein the specified pre-defined mode comprises a parameter defining a receiving width of a workpiece interface of the workholder; and code for causing the workholder to autonomously execute workholding operations pursuant to the parameters of the specified pre-defined mode comprises code for causing the workholder to open the workpiece interface to the receiving width.

20. The non-transitory computer readable medium of claim 16, wherein the specified pre-defined mode comprises: a first parameter defining a receiving width of a workpiece interface of the workholder; and a second parameter defining a clamping width of a workpiece interface of the workholder; and code for causing the workholder to autonomously execute workholding operations

pursuant to the parameters of the specified pre-defined mode comprises: code for causing the workholder to open the workpiece interface to the receiving width; and code for receiving a closing trigger signal subsequent to opening the workpiece interface to the receiving width; and code for causing the workholder to close the workpiece interface to the clamping width in response to receipt of the closing trigger signal.
